

IT8106 Ubiquitous Computing and Intelligent Systems - T2 2026

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IT8106 Ubiquitous Computing and Intelligent Systems - T2 2026

Course code: IT8106 | **Source:** Whitecliffe Canvas (learn.mywhitecliffe.com)

7 modules · 41 content pages · 2 assignments · 48 files.

Course Introduction

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Course Overview

IT8106 Course Overview **Course Description** This course is designed for someone who is looking for specialised knowledge of how computer systems can be infused into the physical world and human and social environments. Content covered in this course is concerned with developing situated and pervasive technology that can be used seamlessly by humans considering the harmony of human and social environments. It will enable you to keep abreast of the latest developments and industry practices. This course will not only enable you to achieve cross disciplinary exchange of ideas with specialised knowledge but will also develop the key skill of design by covering mobile services, sensor networks, context aware computing, intelligent

systems and internet of things. It initially outlines the vision of ubiquitous computing by illustrating real world applications and scenarios and then covers the holistic framework (Smart DEI), which is linked with the core internal properties, external interactions and architectural design of ubiquitous computing systems. It then covers the service provision lifecycle (service announcement, discovery, selection and configuration). As with recent developments of electronic components, these are getting smaller, faster and cheaper. They can be deployed pervasively at massive scale for many industrial applications driven by a need to enrich humans to be better informed about any human computer interactions. The course then covers, tagging, sensing and controlling of smart objects to model context aware systems. It is important to understand that autonomy is one of the core properties of ubiquitous computing systems. This enables systems to operate independently without external interventions and it is a requirement in almost every computing field nowadays that will eventually make IT an invisible component. This course covers different architectures of intelligent systems such as reactive intelligent system model, environment model based intelligent systems, goal based intelligent systems, utility based intelligent systems, learning based intelligent systems, and hybrid intelligent systems. Finally, it covers different communication protocols and an operating system for small, low powered devices to connect them to the internet. Course Level: 8 Course Credits: 15 Course Learning Outcomes On successful completion of this course, you will be able to: LO 1: Apply key ubiquitous computing properties to the architectural design of ubiquitous computing systems. LO 2: Apply service architecture models of sensor technology and sensor networks to model context aware and adaptive systems. LO 3: Evaluate the efficacy of interaction designs from ubiquitous computing systems. LO 4: Justify the design of autonomous systems to support self-operation in dynamic and diverse environments.

Assessment You can find all assessments for this course within the course modules, or by clicking Assignments on the left panel. This will show all assessed tasks. This course has 2 assessments. The assessment due dates will be noted in your course calendar. Assessments in this course will gauge your ability to model the key ubiquitous computing properties of a system in any given scenario. They also measure your ability to utilise and evaluate different interaction design models to meet real-world requirements along with the skills of designing a blueprint for the development of actual systems. In the research and requirements assessment, functional and non-functional requirements analysis is performed that may include, physical, virtual ICT or human environment. The evaluative design solution report includes a ubiquitous computing system architecture based upon the requirement analysis. It covers the design issues, design constraints and choices, and a description of how a part of this solution can be implemented using some mobile device software API. Note this does not mean implementing the application but relating the application design model to specific device APIs, to explain how a main part of the design could be implemented using the APIs / programming. An overview of each assessment is provided here:

Assessment Name	Learning Outcomes	Weighting
Assessment 1: Research and Requirements	LOs 1, 2, 3	30%
Assessment 2: Evaluative Design Solution Report	LOs 2, 3, 4	70%

Disclaimer: All assessment grades are provisional and subject to approval by the School of Information Technology (SoIT) Academic Programme Committee. **Success Criteria** To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed. You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be

awarded 50%. Extensions and Late Submissions Policy An extension can only be granted under exceptional conditions normally for a maximum of (5 working days) , unless there is evidence to support a longer timeframe. Complete and return the following form to your Course Lecturer: [Assessment Extension Request Form.pdf](#) Assessments received one (1) to five (5) working days past the due date, or an approved extended due date, will be marked but will receive a mark capped at the pass mark i.e. 50% for graded assessments. Assessments received more than five (5) working days past the due date, or approved extended due date, will not be marked and will receive a mark of zero or fail. Academic Integrity and Academic Misconduct Policy Whitecliffe is committed to ensuring the highest level of academic integrity and expect staff and students to conduct themselves as responsible and honest members of the academic community. Where there is evidence that breaches of academic integrity may have occurred, the course of Academic Integrity and Academic Misconduct Policy will be followed. This policy considers academic integrity to mean honesty in all matters relating to academic endeavours. Breaches in academic integrity, or academic misconduct, include all forms of cheating incorporates plagiarism (intentional or unintentional), unauthorised collaboration, impersonation, falsification and collusion. Learning Activities Learning activities are provided to develop and consolidate your learning and understanding of the concepts. In preparation for the assessment tasks, make sure you complete each learning activity. You are provided with some reading material, videos, and learning tasks for self-study to support your learning during the course. You are also encouraged to engage with other reliable resources online. The Whitecliffe Library is a helpful place to begin.

Links & References

- [Link] [Assessment Extension Request Form.pdf](#)
- [Link] [Academic Integrity and Academic Misconduct Policy](#)
- [Link] [Whitecliffe Library](#)

Ngā Tikanga Paihere

IT8016 Ngā Tikanga Paihere Icon Progress Bar (browser only) Working in IT requires you to consider the needs and values of the people around you in terms of the systems they interact with. In this course we ask you to integrate Ngā Tikanga Paihere, tikanga Māori, and Te Ao Māori (the worldview principles of the people of New Zealand). Ngā Tikanga Paihere is a framework, originally designed by Statistics NZ, to guide ethical and culturally appropriate data use. It consists of 10 tikanga [Māori world view principles], highlighted below, which can be applied to the IT sector in general. Throughout this course and post graduate, you will see the Ngā Tikanga Paihere framework referenced. Each module will highlight a specific tikanga [principle] relevant to the content covered within the module. A reflection question will also be included at the end of the module to allow you to think holistically about how that principle applies. For more information on the framework and to help you understand the Māori world view better, have a read through the 18-page Ngā Tikanga Paihere Guidelines. Source : [Nga-Tikanga-Paihere-Guidelines-December-2020.pdf](#) (data.govt.nz)

Links & References

- [Link] [tikanga](#)
- [Link] [Ngā Tikanga Paihere](#)

Online Learning Resources

IT8106 Online Learning Resources O'Reilly Learning O'Reilly learning provides individuals, teams, and businesses with expert-created and curated information covering all the areas that will shape our future—including artificial intelligence, operations, data, UX design, finance, leadership, and more. To access O'Reilly Learning, you will need to login using@mywhitecliffe.com email address here: Login to O'Reilly

Links & References

- [Link] [Login to O'Reilly](#)

Turnitin Canvas Plagiarism Framework and Draft Coach

IT8106 What is the Turnitin Canvas Plagiarism Framework Turnitin Canvas Plagiarism Framework The Turnitin Canvas Plagiarism Framework is a way to integrate Turnitin's plagiarism checking service directly into your Canvas assignments. Here are the key features: Similarity Reports This is the core function, allowing lecturers to see how much a student's work matches existing sources. It highlights potential plagiarism and provides an overall similarity score. Lecturers can choose when to make the report visible to students. Simplified Workflow Turnitin works within the Canvas assignment activity. Lecturers can set it up for assignments and view reports during the marking process. The similarity score label provides a link to the similarity report. Students, if allowed, have access to the report from the submission details. Basic Spelling and Grammar Suggestions Turnitin offers basic suggestions for improving a student's writing mechanics. Focus on Originality The framework prioritizes checking for plagiarism without offering Turnitin's full suite of grading functionalities. This keeps the focus on originality. What are the file requirements? Turnitin accepts files that are less than 100MB and less than 800 pages. The allowed filetypes for upload are: Microsoft Word (DOC and DOCX) Portable Document Format (PDF) Plain text (TXT) Rich Text Format (RTF) OpenOffice (ODT) Powerpoint (PPTX) HTML Adobe PostScript Frequently Asked Questions What is Turnitin Canvas Plagiarism Framework? This framework integrates Turnitin's plagiarism checking service directly within your Canvas assignments. It analyses your submitted work against a vast database of sources to identify potential matches. How does it work? Turnitin has been enabled for an assignment. You submit your work through Canvas as usual. Turnitin generates a Similarity Report highlighting potential matches in your work. Your instructor reviews the report alongside your assignment and provides feedback as part of the assessment. What is Turnitin Draft Coach? Turnitin Draft Coach provides you with the ability to use Turnitin's tools while you draft your assignment. You will be able to run a total of three Similarity Checks Links to an external site. . Draft Coach is available as an add-on in Microsoft Word online, accessible through the Student Hub . You can only upload documents with filetypes doc , docx or odt . How to use Turnitin Draft Coach? Login to the Student Hub to an external site. Click on All Apps. In the Apps dashboard, click on Word. Upload your draft assignment and open in Word, or paste your draft text into a blank document. In the ribbon, click on Turnitin and next on Draft Coach. This will open the Turnitin Draft Coach sidebar. You must agree with the Turnitin Terms and Condition when you do a similarity check for the first time. Click on Go to Similarity to run the check. Click on View Full Report to review and refine results. Consult your lecturer for details on the threshold of an acceptable similarity score. How many times can I do Similarity Check for an assignment? You will be able to run a total of three

Similarity Checks while drafting your assignment. We suggest using these strategically to get as much out of the reports as possible. Where can I learn more about Turnitin? Turnitin has a Student hub available, a page with links to the most important guides about Turnitin.

Links & References

- [Link] [Similarity Checks](#)
- [Link] [Student Hub](#)
- [Link] [Student hub](#)

Module 1: Ubiquitous Computing Basics

Module 1: Introduction

Module 1 Ubiquitous Computing Basics: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to this module which introduces you to the basics of ubiquitous computing. Content In this module, you will: Be introduced to the field of ubiquitous computing Explore how the three core designs (smart devices, environments and interaction) can be applied to ubiquitous computing now and in the future Look at what principles and form factors enable ubiquitous computing Investigate a range of interactions between UbiCom devices, people and the physical world Ngā Tikanga Paihere - Wairua The tikanga or principle we will focus on in this module is Wairua , which stands for community good. It's about having clear purpose and action. Wairua can be described as follows: Source: <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf> Reflection As you work through this module, think about any potential harm that could occur when safeguarding privacy. How might ubiquitous computing support community interests and aspirations? How can the voices of the communities of interest be elevated by ubiquitous computing? What mitigation can you put in place to address potential risks or negative impacts created by ubiquitous computing?

Links & References

- [Link] <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf>

What is Ubiquitous Computing?

Module 1 What is Ubiquitous Computing? Icon Progress Bar (browser only) Focus This page introduces the basic concept of Ubiquitous Computing and discusses some related examples. Ubiquitous Computing The term “ ubiquitous ” , meaning appearing or existing everywhere, combined with computing to form the term Ubiquitous Computing (UbiCom), is used to describe ICT (Information and Communication Technology) systems that enable information and tasks to be made available everywhere, and to support intuitive human usage, appearing invisible to the user. Poslad, S. Ubiquitous Computing: Smart Devices, Environments and Interactions Ubiquitous Computing, also known as Pervasive Computing, is a concept where computing is made to appear anytime and everywhere. In contrast to desktop computing, ubiquitous computing can occur anytime using any device and in any location. In a sense, Ubiquitous Computing refers to a vision for computing: To enable computer-based services to be made available everywhere (ubiquitous) To support intuitive human usage But yet, appear

to be invisible to the user Watch the following videos for an introduction to Ubiquitous Computing: Introduction to Ubiquitous Computing What Devices Enable Ubiquitous Computing? Ubiquitous computing devices have evolved to include (but are not limited to): Laptops Notebooks Smartphones Tablets Wearable devices Sensors Everyday Applications The following applications, situated in the human and physical world environments, can illustrate the range of benefits and challenges for ubiquitous computing: A personal memories scenario focuses on the users recording audio-video content, automatically detecting user contexts, and annotating the recordings. Digital cameras automatically capture a visual of the part of the physical world scene on an inbuilt display. This allowed photography to be far less intrusive for the subject than it was previously with the use of film cameras. P.4, 1.1.2.1 in Poslad, S, Ubiquitous Computing: Smart Devices, Environments and Interactions (2009) A scheduled transport service scenario focusing on the transport schedules, adapting their preset plans to the actual status of the environment, and distributing this information more widely. Real-time access to timetables via mobile phones and vehicle GPS, allowing rerouting and rescheduling to be dynamic based on schedules changes, current positions, and the demand for services. P.5, 1.1.2.2 in Poslad, S, Ubiquitous Computing: Smart Devices, Environments and Interactions (2009) A foodstuff management scenario focusing on how analog non-electronic objects such as foodstuffs can be digitally interfaced to a computing system for monitoring their usage. A fully automated foodstuff management system could involve robotics that can move physical objects around and quantify the level of a range of analog objects. One such example is Samsung advertising their new fridge as "The new hub for your home" . This fridge helpfully suggests recipes and meals based on the food inside or your usual shopping preferences, plus many other smart activities. The latest disease detection tool is a smart toilet . But not the kind that lifts its own lid in preparation for use; this toilet is fitted with technology that can detect a range of disease markers in stool and urine, including those of some cancers, such as colorectal or urologic cancers. This technology also has possibilities in the detection of Covid-19. Let's think it over! A relative of yours is totally averse to technology of any kind. Suppose you had to convince them to use a particular device that allows for ubiquitous computing. What arguments would you expect to hear from your relative about using the device? How might you convince them otherwise? How would approaching this problem through the Wairua principle help you to convince your relative? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [Introduction to Ubiquitous Computing](#)
- [Link] ["The new hub for your home"](#)
- [Link] [smart toilet](#)

Core Properties of Ubiquitous Computing

Module 1 Core Properties of Ubiquitous Computing Icon Progress Bar (browser only) Focus This page introduces the core properties of Ubiquitous Computing and discusses some related examples. Some external links to short videos and reading material are also provided. Core Properties Mark Weiser first introduced the vision of ubiquitous computing (or UbiCom) in 1991. He referred to computing as a powerful shift in computation, where people live, work, and

play in a seamless computer-enabled environment, interleaved into the world. Thus, UbiCom postulates a world where people are surrounded by computing devices and a computing infrastructure that supports us in everything we do. The features that distinguish UbiCom from distributed ICT systems are as follows: First, UbiCom systems are situated in human-centred personalised environments, interacting less obtrusively with those around them. Second, UbiCom systems are part of, and used in, physical environments, sensing more of the physical environment. As they are more aware of it, they can adapt to it and are able to act on it and control it. Weiser's vision for ubiquitous computing can be summarised in three core requirements: Computers need to be networked, distributed, and transparently accessible. Human-computer interaction needs to be hidden more. Computers need to be context-aware for optimising their operation in their environment. In addition to the above, there are two additional core types of requirements for UbiCom systems: Computers can operate autonomously, without human intervention, and be self-governed, in contrast to pure human-computer interaction. Computers can handle a multiplicity of dynamic actions and interactions, governed by intelligent decision-making and intelligent organisational interaction. Source: <https://www.slideshare.net/sakshipimple/properties-of-ubiquitous-computing> Additional Resources The following video gives a good overview of Mark Weiser's 1991 predictions and where we are several decades later: The link below will take you to Chapter 1 of Dr Stefan Poslad's book, 'Ubiquitous Computing - Smart Devices, Environments and Interaction.' This will help you develop a deeper understanding of these core properties. Ubiquitous Computing: Smart Devices, Environments and Interactions Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://www.slideshare.net/sakshipimple/properties-of-ubiquitous-computing>
- [Link] [Ubiquitous Computing: Smart Devices, Environments and Interactions](#)

Ubiquitous System Environment Interaction

Module 1 Ubiquitous System Environment Interaction Icon Progress Bar (browser only) Focus This page introduces the UbiCom System Model and discusses some related examples. Some external links to the short videos and reading material are also provided. Ubiquitous System Environment Interaction There are three basic types of environments for UbiCom systems: The infrastructure of other Information and Communications Technology (ICT) systems The Physical World Environment The Human Environment Several basic types of system environment interaction occur: Between Humans and ICT systems Human-Computer Interaction (HCI) Between ICT systems and the physical world Computer-Physical World Interaction (CPI) Between ICT systems C2C or CCI In addition, interactions can occur between the non-ICT systems, such as between different physical world entities and between humans (H2H), also called social interaction. UbiCom System Model The image below shows how core properties of a UbiCom System are associated with different types of environments. Source : <https://www.scinapse.io/papers/187697284> In the image below, you can see how increasing human or computer interaction may result in different types of physical world interactions. For example, increasing human interactions to the maximum and eliminating computer interactions would result in H2H (human to human) interactions. Likewise, increasing machine interactions and reducing human involvement to zero would give rise to C2C

(computer to computer) interactions. A moderate level of human and machine involvement in an interaction may produce H2C (human to computer) or C2H (computer to human) interactions. Source: <https://lekhasuranablog.files.wordpress.com/2020/02/ubicomp-notes-unit1-protected-1-1.pdf> Video Watch Yvonne Rogers' 2014 Ted Talk on how Humans and Technology interact to support human interactions: Reflection Having watched the video, think about: What environments have you experienced personally that promote mindless isolation? How have they made you feel? What environments have you experienced that promote mindful connection? How have they made you feel? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel Concluding Module 1 Well done, tau kē! You have reached the end of this module! What you achieved In this module, you learned the key fundamental concepts needed to understand the working of ubiquitous computing. Coming up The next module will outline how smart devices and services work.

Links & References

- [Link] <https://www.scinapse.io/papers/187697284>

Module 2: Smart Devices and Services

Module 2: Introduction

Module 2 Smart Devices and Services: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to the Smart Devices and Services module, a key part of your ubiquitous computing course. Here, you will explore the technologies that are transforming our everyday lives, from smart homes to intelligent city systems. This module will introduce you to the essential concepts behind smart devices and services, including their design, functionality, and impact on society. You will learn about the various types of smart devices, the role of virtual machines and operating systems, and the core design principles that ensure these technologies are effective and reliable. Additionally, you will study service architecture models that enable smart services to adapt to users' needs. Objectives At the completion of this module, you will be able to: Describe different smart devices and services and their uses. Understand the importance of virtual machines and operating systems in smart technology. Apply design principles for effective and scalable smart systems. Implement service architecture models tailored to users' needs. Assess the societal impact of smart technologies. Ngā Tikanga Paihere - Wānanga The tikanga or principle we will focus on in this module is Wānanga, which stands for organisations. It's about using good data standards and practices. Wānanga can be described as follows: Source: <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf> Reflection As you work through this module, think about how you will work with smart devices and services support ethical and culturally appropriate uses of data. Has ethics been factored in as part of the design? How will the devices and services support iwi and Māori, or other marginalised groups?

Links & References

- [Link] <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf>

Smart Devices and Services Overview

Module 2 Smart Devices and Services Overview Icon Progress Bar (browser only) Focus This page introduces the need for smart devices, their characteristics, and discusses some related examples. External links to short videos and reading material are also provided.

What Are Smart Devices? Smart devices are everyday objects made intelligent with advanced computing, including AI and machine learning, and networked to form the Internet of Things (IoT). They can operate at the edge of the network or on small endpoints, and while they may be small, they are powerful enough to process data without having to report back to the cloud. They range from sensors to refrigerators and wearables to container transportation, capable of running autonomous workloads.

Why Smart Devices? Smart devices can be combined to bring intelligence to both objects and spaces, such as smart homes and buildings, and can help automate processes and controls. They can be used in almost any industry, from smart manufacturing to healthcare, helping to improve efficiency and optimise operations.

<https://www.arm.com/glossary/smart-devices> Smart devices generally connect to other devices or networks through protocols, e.g., Bluetooth, NFC, WiFi, 3G, 4G, etc. For instance, a smartwatch not only tells the time but also measures your heart rate and the number of steps you take in a day.

Smart Device Forms In the 1990s computer scientists started to think about the different forms smart devices might take. By then, microprocessors were getting smaller, cheaper and easier to slot into devices other than desktop computers and ubiquitous computing was gaining traction. In 1991, Mark Weiser proposed three basic forms for ubiquitous system devices: tabs, pads, and boards.

Tabs: Accompanied or wearable centimetre-sized devices, e.g., smartphones, smart cards.

Pads: Hand-held decimetre-sized devices, e.g., laptops.

Boards: Meter-sized interactive display devices, e.g., surface computers and smart boards. These forms are characterised by being macro-sized, having a planar form, and incorporating visual output displays. Three additional forms for ubiquitous systems have since been proposed: smart skins, dust, and clay.

https://en.wikipedia.org/wiki/Smart_device The three additional forms include:

Smart Skins: E.g., fabrics, surfaces.

Dust: Micro electro-mechanical systems embedded in larger devices.

Clay: 3D-printed shapes embedded with smart tabs and dust or combining multiple dust devices into larger ones.

Characteristics The main characteristics of smart devices are as follows:

- Multipurpose:** Can operate as a single portal to access multiple application services, either locally on the device or remotely on servers.
- Personalised:** Usually owned by someone.
- Mobility:** Portable, wearable, or handheld.
- Volatile Service Access:** Dynamic service discovery in various contexts using wireless networks.
- Self-Awareness:** Aware of their physical environment and resources such as memory. Smart devices operate autonomously to some extent without human intervention and can handle multiple dynamic actions and interactions.

Distributed System Viewpoints Smart devices allow for a multitude of views, as illustrated in the image below: Source: <https://slideplayer.com/slide/4804144/>

Smart Devices Today The original form factors of smart devices are still in play, with nearly every household in the West owning at least one smartphone and tablet or laptop. However, there has been significant development in wearable smart technology, such as VR headsets, smart glasses, fitness watches, and healthcare products like smart hearing aids or pacemakers. The smart home is another big area in contemporary smart device usage. This includes Wi-Fi-enabled equipment like smart TV sets, networked fridges, and smart lighting that can be controlled remotely. Manufacturers like Sonoff produce smart light fittings and switches that connect to a secure home Wi-Fi network.

<https://programminginsider.com/what-are-smart-devices/> Activity Read the guide below on smart home technology: 14 Types of Smart Home Technology Options Choose two

options that interest you the most and answer the following questions: Why did you choose this smart technology? What makes this product/technology appealing and beneficial to customers? What design challenges may have impacted it? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://www.arm.com/glossary/smart-devices>
- [Link] https://en.wikipedia.org/wiki/Smart_device
- [Link] <https://slideplayer.com/slide/4804144/>
- [Link] <https://programminginsider.com/what-are-smart-devices/>
- [Link] 14 Types of Smart Home Technology Options

Virtual Machines and Operating Systems

Module 2 Virtual Machines and Operating Systems Icon Progress Bar (browser only) Focus There is a host of potential applications for virtualisation when it comes to smart devices. There is also a need to move beyond the typical operating systems to allow smart devices to thrive. Let's take a look at that next. Virtual Machines and Virtualisation Virtual machines are the fundamental components of virtualisation. They are the containers for traditional operating systems and applications that run on top of a hypervisor on a physical server. Inside a virtual machine, things seem very much like the inside of a physical server—but outside, things are very different. A virtual machine (VM) has many of the same characteristics as a physical server. Like an actual server, a VM supports an operating system and is configured with a set of resources to which the applications running on the VM can request access. Unlike a physical server, where only one operating system runs at any one time, many VMs can run simultaneously inside a single physical server, each supporting different operating systems and applications. In recent times, there has been a lot of interest and discussion about virtualisation from IT professionals. The computing impact in recent years has brought virtualisation back into focus. Virtualisation technology separates computing resources from the operating systems on a physical machine. By emulating complete resource systems, each machine can share a common set of hardware, unaware that this hardware may also be used by another machine simultaneously. This allows multiple machines with different operating systems to run side by side and isolated on the same physical machine. Source: Virtualization for Smart Home Technologies Useful Resources To understand more about the role virtualisation plays in smart devices, check out the article below: Virtualization for Smart Home Technologies Chapter 3: Describing a Virtual Machine (O'Reilly) For more detailed information about Virtual Machines, read Chapter 3 of Virtualising Essentials . Docker vs Virtual Machine Operating Systems Until recently the term Operating System (OS) referred to a small number of possible software solutions. Microsoft Windows dominated most of the world markets, followed by Linux and Mac OS. The list of terminal devices (TD) requiring an OS for their functioning was limited to personal computers, laptops, servers and some gaming consoles. However, in the recent past the number of devices driven by their own OS has been gradually growing, causing a diversification in the field of OSs. New OSs are being developed by either modifying the existing OSs like Android, iOS or Linux, or by creating new software solutions for specific types of devices. Source: Overview: Operating Systems of Modern Terminal Devices Core Reading Read

the 6-page document below for an overview of the operating systems required for smart devices and technologies: Overview: Operating Systems of Modern Terminal Devices Android OS for Smartphones Android is an operating system and programming platform developed by Google for ubiquitous devices, such as mobile phones and tablets. It can run on a variety of mobile devices from different manufacturers. Read more about the Android OS from the site below: Introduction to Android Google Developer Training Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [Source: Virtualization for Smart Home Technologies](#)
- [Link] [Chapter 3: Describing a Virtual Machine \(O'Reilly\)](#)
- [Link] [Docker vs Virtual Machine](#)
- [Link] [Source: Overview: Operating Systems of Modern Terminal Devices](#)
- [Link] [Overview: Operating Systems of Modern Terminal Devices](#)
- [Link] [Introduction to Android](#)
- [Link] [Google Developer Training](#)

Activity: Create a Hello World App using Android

Instructions Open Android fundamentals 01.1: Android Studio and Hello World . Work through the tutorial to: Install the Android Studio development environment. Create an emulator (virtual device) to run your app on your computer. Create and run the Hello World app on the virtual and physical devices. Explore the project layout. Generate and view log messages from your app. Explore the AndroidManifest.xml file. Take a screenshot of your finished app and send it to your lecturer to review. Note : For more detailed information on developing applications using Android Studio, check out the following Google Developer Training - 1.1: Your first Android app .

Links & References

- [Link] [Android fundamentals 01.1: Android Studio and Hello World](#)
- [Link] [1.1: Your first Android app](#)

Core Designs for Ubicom Systems

Module 2 Core Designs for Ubicom Systems Icon Progress Bar (browser only) Focus This page introduces the core designs for ubicom systems and discusses some related examples. Some external links to short videos and reading material are also provided. Core Design Patterns There are three core design patterns for ubiquitous systems (smart devices, smart environment and smart interaction). Take a look at what they are. Source: Poslad, S. "Ubiquitous Computing: Smart Devices, Environments and Interactions" (2009) Check out the table below to compare the differences between a smart device, smart environment and smart interaction Source: Poslad, S. "Ubiquitous Computing: Smart Devices, Environments and Interactions" (2009) Smart Devices and Service Examples See below for examples of devices and services you might find in people's homes. Source : Minchev, Z and Boyanov, L : System modelling and Experimental Assessment of IoT Cyberthreats in Future Smart Homes, Nov 2015

Conference: ICAICTSEE-2015 DOI:10.13140/RG.2.1.1210.0887 General Characteristics The table below sets out the general characteristics of smart devices, environments and interactions. Source : Poslad, S. "Ubiquitous Computing: Smart Devices, Environments and Interactions" (2009) Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [File] [System modelling and Experimental Assessment of IoT Cyberthreats in Future Smart Homes, Nov 2015 Conference: ICAICTSEE-2](#)

Service Architecture Models

Module 2 Service Architecture Models Icon Progress Bar (browser only) Focus This page introduces the service architecture models of ubiquitous systems and discusses some related examples. External links to short videos and reading materials are also provided. N-tier Client Service Model Work through this video to understand more about the N-tier model: N-tier Middleware Read this article to find out how middleware works: What is Middleware and How Does it Work? Service-Oriented Architecture (SOA) Services-oriented architecture Additional Resources If you are interested in going into more detail, you can go through the slides of Service Architecture Models given below: Ubicom-ch03-slides.pdf Source : <http://wiki.icmc.usp.br/images/8/84/Ubicom-ch03-slides.pdf> Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel Concluding Module 2 Well done, tau kē! You have reached the end of this module! What you achieved In this module, you learned about smart devices, services, operating systems and ubicomp systems. Panel Heading t Coming up The next module will outline the ubicom functions of tagging, sensing, and controlling.

Links & References

- [Link] [N-tier](#)
- [Link] [What is Middleware and How Does it Work?](#)
- [Link] [Services-oriented architecture](#)
- [File] [Ubicom-ch03-slides.pdf](#)
- [Link] <http://wiki.icmc.usp.br/images/8/84/Ubicom-ch03-slides.pdf>

Module 3: Tagging, Sensing and Controlling

Module 3 Introduction

Module 3 Tagging, Sensing and Controlling: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, In this module, you will explore the tagging in the physical world, including its applications, its lifecycle, the complexity of tag management, RFID tags and personalised and social tags. You will also look at the key information you need to know about sensors and sensor networks, including Micro Actuation and Sensing (MEMS) and how they work. Ngā Tikanga Paihere - Pūkenga Earlier we introduced the Ngā Tikanga Paihere Guidelines Links to an external site. . The tikanga or principle we will focus on in this module is Pūkenga , which stands for knowledge and expertise. Pūkenga can be described as follows:

Source: <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf> Reflection As you work through this module, think about how you will work with tagging, sensing and controlling in culturally appropriate ways. Who are the people whose data will be recorded? What is their understanding of tagging, sensing and controlling? How do you respect their data in a culturally appropriate way?

Links & References

- [Link] [Ngā Tikanga Paihere Guidelines](#) [Links to an external site.](#)

Tagging, Sensing and Controlling Overview

Module 3 Tagging, Sensing, and Controlling Overview Icon Progress Bar (browser only) Focus This page introduces the concept of tagging, sensing, and controlling in a ubiquitous environment and discusses some related examples. External links to short videos and reading material are also provided. Example: Autonomous Vehicles Let's take an example of Autonomous Vehicles which have made tremendous progress over the last few years. It is important for an autonomous vehicle to monitor the environment continuously for safety and security purposes. To achieve this, vehicles need to have multiple sensors that can collect real-time data on fuel, traffic conditions, weather, speed, components' physical status, and more. Keep this scenario in mind and watch the video below for an overview of tagging, sensing, and controlling. You will learn about the components below as you work through this module. Source : Poslad, S. "Ubiquitous Computing: Smart Devices, Environments and Interactions" (2009) Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Tagging

Module 3 Tagging Icon Progress Bar (browser only) Focus This section explores the 'Tagging' concept, including examples and applications. You will also find links to videos and reading materials for further study. Tagging Overview Physical tags are used to identify, describe, and represent physical objects in a virtual environment as virtual tags. Virtual tags contain metadata or annotations about the physical object, which can be manipulated in the virtual environment. These annotations are linked to the physical tag identifier and may be stored onsite or offsite, e.g., in a database. Tags can be attached to objects to identify different elements in the physical world over a wireless link. Unlike earlier barcode technology, it does not require a line of sight and manual orientation to read the identifying tag. Tagging, therefore, enables ubiquitous computing in the physical world and leads to smart environments. Video: Tagging in the Physical World Take a closer look now at tagging in the physical world by watching the video below. The video explains why physical tags are important for augmented reality and their applications, including an introduction to RFIDs. Watch this video to find out more about the lifecycle for tagging physical objects. Tag Management Physical object annotation faces many management challenges that inhibit their mass consumer use. Tags can be bound to objects at creation or attached later, permanently bound or removable, and may need to be read in challenging environments. Management of these tags is essential, from storage and integration of data to avoiding redundancy and interference from multiple tags and readers. Further Reading Read the following article to learn about the infrastructure concepts and management of tag-based ubiquitous computing applications: Infrastructure-concepts module

4 Tag Management.pdf Source: <https://www.vs.inf.ethz.ch/publ/papers/infrastructure-concepts.pdf> RFID Tags and Applications RFID tags are a type of tracking system that use smart barcodes in order to identify items. RFID is short for “radio frequency identification,” and as such, RFID tags utilise radio frequency technology. These radio waves transmit data from the tag to a reader, which then transmits the information to an RFID computer program. RFID tags are frequently used for merchandise, but they can also be used to track vehicles, pets, and even patients with Alzheimer’s disease. An RFID tag may also be called an RFID chip. <https://www.camcode.com/asset-tags/what-are-rfid-tags/> RFID tags are becoming so ubiquitous in our society that the average person probably encounters them daily without realising it. Did you go to the bookstore today? The book you purchased was probably inventoried using RFID technology. Have you travelled overseas recently? The government tracks travel data using RFID chips in passports. Have you ever had trouble finding a lost pet? Maybe you should consider having an RFID tag implanted in Fluffy to help track her next time she gets lost! Stop the video below at 5:00m as this is where the section on tagging ends. RFID Tagging Applications RFID tagging has many real-world applications. Have a look through the various case studies and RFID applications in the links below. Asset Tracking : Examples of RFID tags used for Asset Management across a variety of industries Education : Examples of RFID tags used for attendance management in schools Textile and Laundry Industry : The integration of RFID in a typical laundry cycle Collecting payments : Many ways to contactless payments during Covid-19 times Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [File] [Infrastructure-concepts module 4 Tag Management.pdf](#)
- [Link] <https://www.vs.inf.ethz.ch/publ/papers/infrastructure-concepts.pdf>
- [Link] <https://www.camcode.com/asset-tags/what-are-rfid-tags/>
- [Link] [Examples of RFID tags used for Asset Management across a variety of industries](#)
- [Link] [Examples of RFID tags used for attendance management in schools](#)
- [Link] [The integration of RFID in a typical laundry cycle](#)
- [Link] [Many ways to contactless payments during Covid-19 times](#)

Sensing

Module 3 Sensing Icon Progress Bar (browser only) Focus This page introduces the concept of sensing and discusses some related examples. External links to short videos and reading materials are also provided. Sensing Sensors are a type of transducer that convert physical phenomena such as heat, light, sound into electrical signals. Sensors often act as an enabler, as inputs to a system behaviour so that it can more favourably adapt. They are often embedded as part of a control loop in pre-programmed systems that perform specialised rather than general-purpose functions, e.g., a temperature sensor may be hard-wired into a heating system. Sensors can be used to do many things, such as monitoring environments, tracking assets through time and space concerning some workflow or process, detecting changes in the environment that are of significance to humans, but which humans are either unable to perceive or are considered too risky, controlling a system with respect to the environment within a defined range of changes, and/or adapting services to improve their utility. Sensors like RFID tags are

networked. The basic architecture is similar: sensors can act as data generators, intermediate or services nodes receive, post-process, and store data, possibly remotely. However, whereas tags generate a fixed electrical signal, sensor data may change because it is the output from a transducer that converts varying physical phenomena into varying signals. Sensors may range in scale from Nano sensors to macro sensors. Sensors come in various formats. Ambient sensors are installed in the environment, e.g., light sensors, temperature, and humidity sensors. Wearable sensors are portable sensors attached to the human body, e.g., motion Sensors, physiological Sensors, and smartphone sensors are embedded in a smartphone device, e.g., GPS, motion sensors. Watch the videos below to find out what sensors and sensor networks are, how they are used and how they work. Start at 5:00m in. The content prior to this covers tagging: Watch the following video to do a deep dive into micro actuation and sensing, or MEMs, as it is called for short. Further Learning Read the following from Mobile Sensors and Context-Aware Computing by Manish J. Gajjar for more detailed information about sensors. Read Chapter 3 to learn about Sensors and Actuators Read Chapter 11 to learn about Sensor application areas

Wireless Sensor Networks (WSN)

A wireless sensor network (WSN) is a wireless network that contains distributed independent sensor devices that are meant to monitor physical or environmental conditions. A WSN consists of a set of connected tiny sensor nodes, which communicate with each other and exchange information and data. These nodes obtain information on the environment such as temperature, pressure, humidity or pollutant, and send this information to a base station. The latter sends the information to a wired network or activates an alarm or an action, depending on the type and magnitude of data monitored. Typical WSN applications include weather and forest monitoring, surveillance, physical monitoring of environmental conditions such as pressure, temperature, vibration, pollutants, or tracing human and animal movements. They use the same transmission medium (which is air) for wireless transmission as wireless local area networks (WLANs). For nodes in a local area network to communicate properly, standard access protocols like IEEE 802.11 are available. However, this and the other protocols cannot be directly applied to WSNs. The major difference is that, unlike devices participating in local area networks, sensors are equipped with a very small source of energy (usually a battery), which drains out very fast. Hence the need arises to design new protocols for MAC that are energy aware. Clearly there is some difference between a traditional WLAN and a WSN, as the latter has limited resources. See below for the typical architecture of a sensor node used in a WSN. Source : Introduction to wireless sensor networks, Mohammad S. Obaidat, Monmouth University, New Jersey, Sudip Misra Publisher: Cambridge University Press

Wireless Sensor Networks are considered to be among the most rapidly evolving technological domains thanks to the numerous benefits that their usage provides. As a result, from their first appearance until the present day, Wireless Sensor Networks have had a continuously growing range of applications. Resources The full article on Wireless Sensor Networks can be accessed [here](#). [Introduction_to_wireless_sensor_networks.pdf](#) Applications of Wireless Sensor Networks Watch the Hardware-Defined Product of the IoT Foundations: Fundamentals course below for more information on sensors. Android OS and Smartphone Sensors Most smartphones have built-in sensors that measure motion, orientation, and various environmental conditions. These sensors provide raw data with high precision and accuracy. Explore the links below to learn more about specific sensors supported on Android platforms. Sensors and Android Operating systems How Android OS helps in collecting data from built-in smartphone sensors How to access smartphone motion sensor data in Android OS Sensors: programming mobile applications for Android Handheld systems Watch this video to

understand how you can access and manage the device sensors using Android Smart Sensors and Applications A smart sensor or an intelligent sensor is a device that has integrated electronics, and they are able to perform certain functions like, logic functions, two-way communications, and are capable to make decisions. These sensors are capable to measure the environmental data more accurately with less noise. Smart sensors are used for monitoring and to do the controlling mechanism in many industrial applications. Mostly a smart sensor has three important parts they are sensing element, signal processing part, and microprocessor. The major difference between smart sensors and a conventional sensor is that the smart sensor is faster than the conventional type and it is accurate too. Smart sensors are smaller in size than a normal sensor and they also consume low power. Watch the video below to gain a good understanding of Smart Sensors and their different types: Smart Sensors and Industry 4.0 In Industry 4.0, physical processes are seamlessly integrated with software and computational processes into “cyber-physical systems”. Such systems typically involve intelligent sensing technologies at the data measurement and monitoring level of the system architecture. Smart sensors consist of three elements that convert input from the physical environment into analysable data: sensor, microprocessor, and communications unit. While the values measured by more traditional sensors are interpreted entirely by humans, smart sensors are frequently connected with each other via networks and use algorithms to analyse data. They are enhanced with soft sensors that do not directly measure but instead use software modules to process secondary collected data. Smart sensors are an immensely important component of Industry 4.0. Together with special software and additional technical actuators, they are what enables the responsive and agile production that characterizes Industry 4.0. Source : https://www.ey.com/en_ch/industrial-products/how-smart-sensors-are-driving-industry-4-0-forward Further Reading Read more from the links below about how smart sensors are impacting industry. How Smart Sensors are driving Industry 4.0 forward : This article gives an overview of sensors' impact on Industry 4.0. Sensors as drivers of Industry 4.0 : This comprehensive report outlines the impact of smart sensors in Germany, Switzerland, and Austria. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Videos

- <https://www.youtube.com/watch?v=fGr7IbJnFlw>
- <https://www.youtube.com/watch?v=IU-EAtITRRM>

Links & References

- [Link] [Mobile Sensors and Context-Aware Computing](#)
- [Link] [Read Chapter 3 to learn about Sensors and Actuators](#)
- [Link] [Read Chapter 11 to learn about Sensor application areas](#)
- [File] [Introduction_to_wireless_sensor_networks.pdf](#)
- [File] [Applications of Wireless Sensor Networks](#)
- [Link] [Sensors and Android Operating systems](#)
- [Link] [How Android OS helps in collecting data from built-in smartphone sensors](#)
- [Link] [How to access smartphone motion sensor data in Android OS](#)
- [Link] [Sensors: programming mobile applications for Android Handheld systems](#)

- [Link] <https://www.linkedin.com/learning/java-design-patterns-and-apis-for-android/manage-device-sensors?u=36492252>
- [Link] https://www.ey.com/en_ch/industrial-products/how-smart-sensors-are-driving-industry-4-0-forward
- [Link] [How Smart Sensors are driving Industry 4.0 forward](#)
- [File] [Sensors as drivers of Industry 4.0](#)

Controlling

Module 3 Controlling Icon Progress Bar (browser only) Focus This page introduces the concept of 'controlling' and discusses some related examples. Additionally, some external links to short videos and reading material are provided. Introduction To understand the concept of 'controlling', we will start with the definitions of embedded vs real-time systems and then will watch a short video to clarify our understanding. An embedded system is a computer whose end function is not to be a computer. For example, a microwave oven contains a computer, but obviously it's end function is to heat up food. A television contains a computer, but it's end function is to display audio and video. A real-time system responds in a (timely) predictable way to unpredictable external stimuli arrivals. In short, a Real-Time System has to fulfill under extreme load conditions: Timeliness: meet deadlines, it is required that the application has to finish certain tasks within the time boundaries it has to respect Simultaneity or simultaneous processing; more than one event may happen simultaneously, all deadlines should be met Predictability: the real-time system has to react to all possible events in a predictable way Dependability or trustworthiness: it is necessary that the real-time system environment can rely on it. <https://www.slideserve.com/velvet/examples-of-embedded-and-real-time-systems> Video Resources Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://www.slideserve.com/velvet/examples-of-embedded-and-real-time-systems>

Modelling Context-Aware Systems

Module 3 Modelling Context-Aware Systems Icon Progress Bar (browser only) Focus After discussing the concepts of Tagging, Sensing, and Controlling, this module will cover how these concepts can help in modelling a context-aware system. Some external links to short videos and reading material are also provided. Modelling Context-Aware Systems Context-awareness is considered one of the fundamental properties of UbiCom systems and is a key property of smart environments. Generally, a context represents the state or situation in the environment of a system that affects that system's behaviour. Context-aware systems are systems that are aware of their situation (or context) in their physical, virtual (ICT), and user environment and can adapt the system to this in some way, benefiting from the knowledge of that situation. Context-aware Computing: This aims at adapting the application or the computer in accordance with its changing context. "Context of the application" includes information about who is using the computer; who else is nearby; ambient information about the room, including light, sound, and temperature; physical materials and tools used; and other devices in a room. For example,

a context-aware hospital bed with embedded computers, displays, and sensors can react and adapt to what is happening in its proximity; it may recognise the patient in the bed; it may recognise clinical personnel approaching; it may bring up relevant medical information on the display for the clinicians, and it may issue a warning if the nurse is mistakenly trying to give the patient another patient's medication. Additional Resources YouTube For a detailed explanation of context-aware computing and smart environments, watch the recorded webinar session below. The link will take you to YouTube, as this video can only be played on that platform. Context-aware computing and the Smart environment O'Reilly Read Chapter 2 in the course textbook to find out more about Context-Aware Computing. Context-Aware Computing Types of Contexts There are several dimensions to characterise contexts, and there are many definitions of how to make systems aware of changes in their context (context awareness). A few significant types of context (context awareness) are given below. Physical Environment Context What: Awareness of the type of physical environment or physical phenomena such as awareness of temperature, light intensity, chemical or biological concentration, etc. Where: Spatial awareness or location awareness that may be specified in terms of the current location, or one location concerning another location. When: Temporal awareness that specifies when context awareness is helpful. Is it now, later, or during some activity? It may be set in terms of an absolute time or a relative time to some other event or condition. ICT Environment Context How: Awareness of how any context is created and adapted over an ICT infrastructure, e.g., a context or context-aware application can be accessed over a wireless link and via a mobile terminal. User Environment Context Who: Knowing who might get affected by the awareness of someone's context, either in a personal or social capacity. Context-Aware System Architecture The image below shows the detailed architecture and a model for context-aware systems, which consists of four main components: Current context capture Goal context creation Adaptation of the current context to the goal context Context management The core design feature of this architectural model is that it articulates in detail the distribution and interaction of four main groups of context-based functions, including user/environment context creation, context processing, and context maintenance, which form a life cycle process for operating a context-aware system. The context-aware operational life cycle looks as follows: The physical, human, and ICT environment context Data is created The user goal context needs are created The environment contexts are used to adapt some user or application goal context The context data needs to be managed Source : Poslad, S. "Ubiquitous Computing: Smart Devices, Environments and Interactions" (2009) Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel Concluding Module 3 Well done, tau kē! You have reached the end of this module! What you achieved In this module, you took a look at the major functions of systems in ubicom - tagging, sensing, and controlling. Coming up In the next module, you will consider the intelligent systems.

Links & References

- [Link] [Context-aware computing and the Smart environment](#)
- [Link] [Context-Aware Computing](#)

Module 4: Intelligent Systems

Module 4 Introduction

Module 4 Intelligent Systems: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to this module on Intelligent Systems. Intelligent systems are technologically advanced machines that can perceive and respond to what's happening around them. They can take many forms, including automated vacuums, facial recognition programs, personalised shopping suggestions. At the end of this module, you will be able to: Develop an appreciation for intelligence systems and explain how intelligent agents work in different systems. Learn how knowledge-based systems (KBS) work and explain their architecture. Ngā Tikanga Paihere - Tika The tikanga or principle we will focus on in this module is Tika, which stands for value for all. It's about maintaining public confidence and trust to use data. Tika can be described as follows: Source:

<https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf> Reflection As you work through this module, think about what intelligent systems are attempting to do and how they will benefit the people they serve. What positive effects do you anticipate from intelligent systems? How do intelligent systems contribute towards the long-term wellbeing of our people?

Links & References

- [Link] <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf>

Intelligent Systems Overview

Module 4 Intelligent Systems Overview Icon Progress Bar (browser only) Focus This page introduces the idea of intelligent systems and discusses the applications and challenges of intelligent systems. Some external links to short videos and reading material are also provided. Intelligent systems Intelligent systems (IS) use artificial intelligence (AI), also referred to as machine intelligence, computational intelligence, and include intelligent agent-based systems, software agents, and robots. AI is the ability of a digital computer, machine, or computer-controlled robot to perform tasks commonly associated with intelligent beings. It centres on building intelligent machines that are capable of performing tasks that generally require human intelligence, e.g., the ability to reason, discover meaning, generalise, or learn from past experience. However, with intelligent systems, these tasks can be carried out without any human intervention. Check out the video below where Michael Black teaches systems to recognise people as such. As systems become more intelligent and complex, computers need to be able to figure out by themselves the value of the data they have collected so they can make good decisions. Watch Christian Møller share his insights on how we create truly intelligent systems. Applications of Intelligent Systems Intelligent systems are poised to fill a growing number of roles in today's society, including: Factory automation Field and service robotics Assistive robotics Military applications Medical care Education Entertainment Visual inspection Character recognition Human identification using various biometric modalities (e.g., face, fingerprint, iris, hand) Visual surveillance Intelligent transportation Challenges in Intelligent Systems Research in intelligent systems faces numerous challenges, many of which relate to representing a dynamic physical world computationally. Uncertainty : Physical

sensors/actuators provide limited, noisy and inaccurate information/action. Therefore, any actions the system takes may be incorrect both due to noise in the sensors and due to the limitations in executing those actions. Dynamic world : The physical world changes continuously, requiring that decisions be made at fast time scales to accommodate for the changes in the environment. Time-consuming computation : Searching for the optimal path to a goal requires extensive search through a very large state space, which is computationally expensive. The drawback of spending too much time on computation is that the world may change in the meantime, thus rendering the computed plan obsolete. Mapping : A lot of information is lost in the transformation from the 3D world to the 2D world. Computer vision must deal with challenges including changes in perspective, lighting and scale; background clutter or motion; and grouping items with intra/inter-class variation.

<https://www.unr.edu/cse/undergraduates/prospective-students/what-are-intelligent-systems>
Additional Reading Read the content of the chapter given below to develop your concepts further: [Introducing Intelligent Systems | Building Intelligent Systems: A Guide to Machine Learning Engineering \(oreilly.com\)](#) Activity Reflect on the following questions: What is your favourite hobby that would benefit from an Intelligent System? What is an example of a common activity you do every day where an Intelligent System might not be right? Why not? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? [IT8106 Support and Feedback Channel](#)

Videos

- <https://www.youtube.com/watch?v=FuaLGGAFDIA>

Links & References

- [Link] <https://www.unr.edu/cse/undergraduates/prospective-students/what-are-intelligent-systems>
- [Link] [Introducing Intelligent Systems | Building Intelligent Systems: A Guide to Machine Learning Engineering \(oreilly.com\)](#)

Intelligent System Agents

Module 4 Intelligent System Agents Icon Progress Bar (browser only) Focus This page introduces the details of intelligent system agents and discusses the applications through some external links to short videos and reading materials. Intelligent System Agents An IS system is based upon a conventional distributed system architecture with three enhancements: Supports richer models of interaction between IS and their environments. Uses processing models that support advanced system cognition and behaviours such as searching, reasoning, planning, and learning. Uses a richer communication model with other intelligent systems and intelligent environments. An environment in AI is the surrounding of the agent. An agent generally refers to an autonomous entity that acts upon an environment on behalf of human beings, using observation through sensors and consequent actuators. The agent takes input from the environment through sensors and delivers the output to the environment through actuators. There are several types of environments for intelligent systems in which an agent can work. Watch the video below [29m 52s] for examples of task environments and characteristics to help you understand intelligent agents and their types better. Core Reading A detailed explanation regarding the different types of intelligent agents and IS architecture models can be found at

the links below, including simple reflex agents, model-based reflex agents, goal-based agents, utility-based agents, and learning agents. Intelligent Agents in Artificial Intelligence Intelligent Agents Publication - Intelligent Agents: A Comprehensive Survey Intelligent Agents: Issues and Applications Optional Reading For a detailed study on intelligent agents, their types, and environments, read Chapter 2 of the book “ Artificial Intelligence - A Modern Approach ” by Stuart J. Russell and Peter Norvig. Intelligent Agents Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [Intelligent Agents in Artificial Intelligence](#)
- [Link] [Intelligent Agents](#)
- [File] [Publication - Intelligent Agents: A Comprehensive Survey](#)
- [Link] [Intelligent Agents: Issues and Applications](#)
- [File] [Intelligent Agents](#)

Knowledge-Based Systems (KBS)

Module 4 Knowledge-Based Systems (KBS) Icon Progress Bar (browser only) Focus This page introduces knowledge-based systems and discusses the applications and challenges of knowledge-based systems. Some external links to short videos and reading material are also provided. Knowledge-Based Systems (KBS) Source:

<https://in.pinterest.com/pin/779685754236947810/> Links to an external site. In the last decade or so, expert systems also called knowledge-based systems have made their way from research laboratories into the real world. Applications have been, and are continuing to be, developed in areas as diverse as business, medicine, manufacturing, defence, astronomy, science and engineering. Such applications perform tasks that include interpretation, prediction, diagnosis, design, planning, monitoring, debugging, repairing, instruction and control. The expert systems are the offshoots of artificial intelligence which is concerned with using computers to simulate human intelligence in a limited way. Some researchers define artificial intelligence (AI) as the science of making machines do things that would require intelligence if done by men. In the last few decades, AI has spread into major subfields including expert systems, artificial neural networks, fuzzy systems, evolutionary computation and chaos theory. Some researchers do not differentiate between expert systems and knowledge-based systems. The key issue behind all these developments is the knowledge acquisition, knowledge representation and knowledge processing <https://ieeexplore.ieee.org/document/403493> an external site. The typical architecture of a knowledge-based system, which informs its problem-solving method, includes a knowledge base and an inference engine. The knowledge base contains a collection of information in a given field. The inference engine deduces insights from the information housed in the knowledge base. KBS also includes an interface through which users query the system and interact with it. A KBS may vary with respect to its problem-solving method or approach. Some systems encode expert knowledge as rules and are therefore referred to as rule-based systems. Another approach, case-based reasoning, substitutes cases for rules. Cases are essentially solutions to existing problems that a case-based system will attempt to apply to a new problem. Core Reading Have a read through the information on the site below to find out more about knowledge-based systems, how they have evolved, common features, benefits, and

how they work. Create a Knowledge Base to Stimulate Information Sharing and Boost Organizational Value Reflection Have you ever worked with a knowledge-based system? If yes: What was it? How did it work? How was it beneficial? If you have not worked with a knowledge-based system before, do some research on what systems are available. Try reading some related research articles from the journal mentioned below. Knowledge-Based Systems | Vol 266, 22 April 2023 | ScienceDirect.com by Elsevier Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://in.pinterest.com/pin/779685754236947810/>
- [Link] <https://ieeexplore.ieee.org/document/403493>
- [Link] [Create a Knowledge Base to Stimulate Information Sharing and Boost Organizational Value](#)
- [Link] [Knowledge-Based Systems | Vol 266, 22 April 2023 | ScienceDirect.com by Elsevier](#)

Knowledge Representation

Module 4 Knowledge Representation Icon Progress Bar (browser only) Focus This page introduces the details of knowledge representation and discusses the applications through some external links to short videos and reading materials. Watch and Learn Watch the Edureka video below to find out what knowledge representation is and how reasoning works in AI. The video explains the different techniques that can be used to represent knowledge. Experiment Let's play an experiment to give you a better understanding of how logical reasoning can be related. The Wason Selection Task 1 (philosophyexperiments.com) Further Reading For a detailed study of knowledge-based systems, refer to Knowledge-Based Systems by Akerkar. Pay particular attention to the following sections: Chapter 1, Sections 1.5 to 1.9 Knowledge-Based Systems Objectives of KBS Components of KBS Categories of KBS Difficulties with KBS Activity: Key Components of an Expert System For this activity, start by watching the video: What is an Expert System? Identify the major components that make up an expert system. Provide a short explanation of each component and how it works. Send your work to your lecturer to review, when you are done. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel Concluding Module 4 Well done, tau kē! You have reached the end of this module! What you achieved In this module, you considered what intelligent system - and what makes them intelligent! Coming up The next module scrutinises autonomous systems and artificial life.

Links & References

- [Link] [The Wason Selection Task 1 \(philosophyexperiments.com\)](#)
- [Link] [Knowledge-Based Systems](#)
- [Link] [What is an Expert System?](#)

Module 5: Autonomous Systems and Artificial Life

Module 5 Introduction

Module 5 Autonomous Systems and Artificial Life: Overview Icon Progress Bar (browser only)

Kia ora koutou, hello everyone, Welcome to this module on autonomous systems, complex systems and artificial life. In this module, you will: Explore autonomous or self-managing systems, including the make-up of their core components and how these lead to higher-order self-maintenance properties of the group as a whole Investigate how reflection and self-awareness are inspiring autonomic thinking Look at how complex systems and artificial life are informed by natural and biological processes and phenomena, so they can scale and be resilient to changes in the environment. Ngā Tikanga Paihere - Mauri The tikanga or principle we will focus on in this module is Mauri , which stands for provenance and transformation. It's about having clear purpose and action. Mauri can be described as follows: Source:

<https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf> Reflection As you work through this module, think about how autonomous systems can add value when managed well. How will data be transformed from its original collection purpose? How will data enhance the system?

Links & References

- [Link] <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf>

Autonomous System Overview

Module 5 Autonomous Systems Icon Progress Bar (browser only) Focus This page introduces

the concept of Autonomous Systems and discusses some related examples. Autonomous Systems Overview An autonomous system is one that can achieve a given set of goals in a changing environment – gathering information about the environment and working for an extended period of time without human control or intervention. Driverless cars and autonomous mobile robots (AMRs) used in warehouses are two common examples. Autonomy requires that the system be able to do the following: Sense the environment and keep track of the system's current state and location Perceive and understand disparate data sources Determine what action to take next and make a plan Act only when it is safe to do so, avoiding situations that pose a risk to human safety, property, or the autonomous system itself An autonomous system can sense, perceive, plan, and act without intervention.

<https://blackberry.qnx.com/en/autonomous-systems/> Autonomous vs. Semi-Autonomous Systems The vast majority of current systems known as autonomous are semi-autonomous, rather than fully autonomous. Cars with driver support systems such as lane keep assist and advanced braking systems are semi-autonomous, as are robotic surgery systems, robot vacuums, and most unmanned aerial vehicles (UAVs and drones). Most fully autonomous systems, such as driverless cars, remain too costly, data-intensive, power-consuming, or unsafe for widespread public use. Significant progress is being made. Electric vehicles, autonomous vehicle perception systems, and connected car platforms have passed Gartner's "peak of inflated expectations" and through the "trough of disillusionment." These systems have reached the "slope of enlightenment" in the Connected Vehicle and Smart Mobility Hype Cycle . <https://blackberry.qnx.com/en/autonomous-systems/> Examples of Autonomous Systems

Examples of autonomous systems include autonomous vehicles, autonomous robots, autonomous warehouse and factory systems, and autonomous drones. You'll explore examples in more detail shortly. The Future of Autonomous Systems Check out the video below where Gurdeep Pall, CVP of Business AI at Microsoft, shares his views on the positive impact that autonomous systems will have on our planet, economy, and lives. Reflection After watching the video: What are your thoughts on autonomous systems? What autonomous system innovations excite you? Core Reading Refer to the following PDF from IM=BM which gives an interesting insight into strategies and deployment considerations for manufacturers when it comes to autonomous systems. How to partner your people with AI: Strategies and deployment considerations for manufacturer Activity Well done on reaching the end of this page. So far, this section covers the concept of Autonomous Systems with basic real-life examples. Choose two examples of autonomous systems. Do some research to find out more about each chosen example. Write a paragraph that summarises your findings for each example and share in the discussion forum. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://blackberry.qnx.com/en/autonomous-systems/>
- [Link] [Connected Vehicle and Smart Mobility Hype Cycle](#)
- [Link] [How to partner your people with AI: Strategies and deployment considerations for manufacturer](#)

Autonomous System Pattern

Module 5 Autonomous System Pattern Icon Progress Bar (browser only) Focus This page introduces the concept of Autonomous Systems and discusses some related examples. One of the ultimate goals of artificial intelligence is the ability for machines to operate on their own, with little or any human interaction. This idea of autonomous systems makes up one of the seven patterns of AI Links to an external si that represents the common ways that organizations are applying AI. While some of the patterns are focused on predictive analytics or conversational patterns, or systems that can recognize things in the world around us, those patterns still involve human interaction. After all, we need humans to be involved in conversational or recognition systems. However, the autonomous pattern is much more complicated as we're asking a machine to do something in the real world without a human in the loop. These sorts of systems are harder to implement and generally take longer to show ROI. The Autonomous Pattern Explained Autonomous systems are defined as systems that are able to accomplish a task, achieve a goal, or interact with their surroundings with minimal to no human involvement. It is also essential that these systems be able to predict, plan, and be aware of the world around them. This applies both to physical hardware autonomous systems as well as software autonomous systems (software "bots"). Examples of this pattern include autonomous vehicles and machines, and autonomous bots of all sorts. We also see autonomous systems in the form of autonomous documentation and autonomous knowledge generation as well as autonomous processes and cognitive automation. Autonomous systems have the ability to create legal and medical documentation, invoices, and automatically log data. It can also help businesses automatically route tickets or workflows. With its ability to help with inventory

forecasting, improve shipping times, and tracking, it's easy to see why it's talked about in a positive light around the area of logistics. Collaborative bots, or cobots, are also a form of an autonomous system. Cobots are meant to operate in conjunction with, and in close proximity to, humans to perform their tasks. This is in contrast to industrial robots that are caged off and physically separated from humans. Even though cobots can operate in an augmented intelligence role, they're meant to operate independently of humans even though they're in close proximity. The primary objective of the autonomous systems pattern is to minimize or eliminate human labour. When removing a human from the equation, you need to make sure that the autonomous system is as close to human-level performance as possible. This is why this pattern is one of the harder ones to implement. Naturally, as autonomous systems aim to minimize human labour, they need to be reliable, consistent, and of a very high standard.

Levels of Autonomy When many people think of examples in the autonomous pattern, they instantly think of autonomous vehicles. To have a vehicle that fully drives a human from point A to point B safely without the human ever having to get involved or take over driving would be incredible and is something we can all imagine. However, as we move from vehicles that are fully operated by humans to fully autonomous, it's not an all-or-nothing approach. The Society for Automotive Engineers has defined six levels of autonomy from Level 0 to Level 5. At one end of the spectrum, level 0 autonomous vehicles require full human control of all operation and decision making. At the other end of the spectrum, level 5 fully autonomous vehicles will bring dramatic changes to most industries and will have major societal and economic impacts. Levels of autonomy are necessary due to the high-risk nature of real-life autonomous applications. There is little room for error so further precaution must be taken with each level. Incorporating autonomous systems into a project or real-life situation is a complex and difficult task and autonomous applications need to be as close to perfect as possible. When people think of the autonomous pattern, they also think of robots such as Rosie from the Jetsons, C3PO or R2D2 from Star Wars, or other robots coming from Hollywood. These robots are able to perform many tasks that humans can, such as talking, cooking, picking up objects, avoiding obstacles in front of them, and an array of other things that humans do. However, in reality, many robots have varying levels of intelligence, if any. Industrial robots, for example, are simply machines that are programmed to perform a repetitive task. The vast majority of robotic systems contain no machine learning system or intelligence so to speak. These robots are programmed to complete repetitive and laborious tasks with the aim of decreasing human labour while performing various tasks more efficiently. In essence, this is automation, allowing for a process or procedure to be performed with minimal human assistance, but containing no intelligent features. The oft-overlooked fact is that automation does not correlate to intelligence. Just because a system can repeat something that a human instructed it to do does not mean it has the capacity to learn anything. These machines are simply programmed to perform the same task over and over again. The misunderstanding of automation as being something more than just human instructions and programming could be due to the general public's tendency to envision the version of robots that the media portrays upon the mention of robotics.

Software Automation This leads to the discussion of software automation as well as hardware. The popularity of Robotic Process Automation (RPA) is real. People want to be able to automate software-based processes that ordinarily require a lot of human activity. Despite RPA's lack of intelligent processes, the tasks they perform are vital in helping companies to ameliorate their cognitive intelligence systems. Moreover, these machine processes significantly lower labour costs and can show immediate ROI despite their lack of AI. Indeed, many consider RPA a

gateway to AI. RPA does not have to remain strictly AI-free though. Just as AI can be added to industrial robots, it can also be added to RPA. An asset of this collaboration would be the ability to create intelligent workflow automation that could automatically detect changes in software or systems and work bottlenecks. Along the same vein runs the possibility of developing autonomous business processes that would allow for the optimisation and analysis of business processes without the need for humans. Over the past several months, a few of the large RPA vendors are indeed looking to incorporate AI into RPA. While the autonomous patterns system of AI is arguably the hardest to implement, when done correctly, it can have incredible impact. It is recommended that much thought be put into any step that might involve autonomy. With the many possibilities, the future of the autonomous pattern is a bright one.

[https://www.forbes.com/sites/cognitiveworld/2020/05/30/the-autonomous-systems-pattern-](https://www.forbes.com/sites/cognitiveworld/2020/05/30/the-autonomous-systems-pattern-of-ai/?sh=431a5efa6a6b)

[of-ai/?sh=431a5efa6a6b](https://www.forbes.com/sites/cognitiveworld/2020/05/30/the-autonomous-systems-pattern-of-ai/?sh=431a5efa6a6b) To better understand autonomous systems patterns, let's explore a real-life example: Autonomous Vehicles. Autonomous Vehicles Autonomous vehicles, commonly known as self-driving cars, are an excellent illustration of autonomous systems patterns in action. These vehicles use a combination of sensors, algorithms, and control systems to navigate and drive safely without human intervention. Here is how various autonomous systems patterns are applied: Perception: Autonomous vehicles are equipped with a variety of sensors, such as lidar, radar, cameras, and ultrasonic sensors, to perceive their surroundings. These sensors continuously collect data on road conditions, nearby objects, pedestrians, and other vehicles. Decision-Making: Advanced AI algorithms process the data from sensors to make real-time decisions. Decision-making algorithms determine the vehicle's speed, lane changes, braking, and other driving actions. These algorithms consider factors like traffic rules, road signs, and the behaviour of nearby vehicles. Control: Control systems manage the vehicle's actuators, including the engine, brakes, and steering. They ensure that the vehicle executes the decisions made by the AI safely and accurately. Learning and Adaptation: Autonomous vehicles continually learn from their experiences and data collected during driving. Machine learning models improve their performance over time, adapting to various driving conditions and scenarios. Safety and Reliability: Safety features such as redundant sensors and fail-safes are integrated into autonomous vehicles to reduce the risk of accidents. Algorithms prioritise safety, such as avoiding collisions and following traffic laws. Human Interaction: Many autonomous vehicles include user interfaces that allow passengers to monitor the vehicle's status and intervene when necessary, although full autonomy is the goal. Communication: Autonomous vehicles can communicate with each other and with infrastructure systems (V2V and V2I communication) to enhance safety and traffic efficiency. Scalability and Modularity: Autonomous vehicle systems are designed with scalability in mind, allowing for the integration of new hardware or software updates as technology evolves. Ethical Considerations: Autonomous vehicles raise ethical questions, such as how they should prioritise decisions in the event of unavoidable accidents (the "trolley problem") and issues related to data privacy. In the context of autonomous vehicles, these patterns collectively enable a car to safely navigate through various environments, make complex decisions, and adapt to changing conditions without human intervention. Companies like Waymo, Tesla, and others have been at the forefront of developing autonomous vehicle technology, utilising these patterns to create increasingly autonomous and safe driving experiences for the future. Types of Autonomous Systems There are four major types of design for more general autonomous systems: Design and Interface Autonomy: Autonomous systems can be designed to consist of dynamically reusable and extensible components to enable new tasks to be assigned to existing components.

Simpler autonomous systems can be composed into more complex systems by defining interfaces. This type of autonomy is referred to as design and interface autonomy. Event-Driven Architectures (EDA) and Context-Aware System Designs: Autonomous systems that operate in dynamic environments need to be aware of how dynamic environment states change context-awareness. These systems can be designed to sense events, filter them, select actions, and also modify this selection if the context changes (context adaptation). Hybrid Goal-Based and Environment-Model-Based IS Systems: Rather than specifying a specific set or sequence of actions for a system to do which can easily fail if any one of the actions fails, goal-based planning systems can dynamically plan or self-plan their own actions to achieve a goal, enabling users to specify high-level goals for an IS. Pre-Configured Systems: Autonomous systems can be designed to be pre-configured with inbuilt local goals to define their execution that may or may not meet some global constraints. Once their operation starts, they are self-regulating without any global control. Further Reading Refer to the following web pages to learn about autonomous systems: When Autonomous Systems Meet Accuracy and Transferability through AI: A Survey Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [seven patterns of AI](#)
- [Link] [Collaborative bots, or cobots](#)
- [Link] [augmented intelligence](#)
- [Link] [autonomous systems pattern](#)
- [Link] [The Society for Automotive Engineers](#)
- [Link] [automation does not correlate to intelligence](#)
- [Link] <https://www.forbes.com/sites/cognitiveworld/2020/05/30/the-autonomous-systems-pattern-of-ai/?sh=431a5efa6a6b>
- [Link] [When Autonomous Systems Meet Accuracy and Transferability through AI: A Survey](#)

Reflective and Self-Aware Systems

Module 5 Reflective and Self-Aware Systems Icon Progress Bar (browser only) Focus This page introduces the concept of Reflective and Self-Aware Systems and discusses some related examples. Context and Explanation Context-awareness focuses on an awareness of a system's external environment, the user, the physical environment, and the ICT infrastructure. It periodically senses this environment, automatically detects significant changes, and uses this information to adapt the system's internal behaviour accordingly. An important design issue is determining the degree, if any, of awareness the system needs of its own internal behaviour. The basic design of an ICT system is that it does not process or is unaware of its own actions. However, an intelligent (and self-aware) machine knows what it is doing and what to do next. Much of the focus in a self-aware system is on recognising what the environment is doing and selecting the appropriate action. Reflection is the process by which a system can observe its own structure and behaviour, reason about its behaviour, and possibly modify its behaviour to achieve the desired goals. There are three elements to the reflection process: Exposure and observation of the internal behaviour to support observation (instrumentation) Reflective

computation about the operational computation (introspection) Possible modification to the operational computation (adaptation) Self-Aware Examples A simple application of self-awareness is to monitor the use of online resources to check their status and modify the internal behavior to conserve resources more as resources reach minimal levels. For example, such a system could decide to increase the compression at which it stores content because it assesses that the recording of the current program may exceed the storage to complete the recording in uncompressed mode. Another core application of self-awareness is robots, e.g., robot arms must be aware of their inertial force and braking force in order to move and position accurately in the physical world. More complex self-awareness can involve re-planning actions through analyzing and predicting any current plans of actions that will likely fail. Reflection has several benefits for UbiCom, such as: Facilitating debugging of operational errors in systems as a precursor to improved system fault detection and autonomy Improving (external) context-awareness by supplementing it with a system's self-awareness Enabling greater system adaptation of run time system behavior possibly as a precursor to fault correction and supporting self-explanation to external environments. Source: " Ubiquitous Computing: Smart Devices, Environments and Interactions " by Stefan Poslad , John Wiley & Sons, Ltd. ISBN: 978-0-470-03560-3 Case Study: The Need for Reflective and Self-Aware Systems in Healthcare Background: In the healthcare industry, reflective and self-aware systems become crucial when considering the challenges associated with medical diagnosis and treatment decisions. Case Scenario: Sarah, a 35-year-old woman, has been suffering from unexplained symptoms such as chronic fatigue, joint pain, and digestive issues. Despite numerous tests, her condition remains undiagnosed, and her health continues to deteriorate. The Problem: Data Overload: Sarah's case has generated an overwhelming amount of medical data, making it difficult for healthcare professionals to arrive at a conclusive diagnosis. Diagnostic Errors: Healthcare providers may face cognitive biases or knowledge limitations that can lead to diagnostic errors or delays in treatment. The Solution: A reflective and self-aware diagnostic support system is employed to assist healthcare professionals in diagnosing Sarah's condition. Key Features of the System: Data Integration: The system collects and integrates all of Sarah's medical data. Knowledge Base: It has access to an extensive medical knowledge base, including the latest research findings and clinical guidelines. Machine Learning Algorithms: The system uses machine learning to analyse Sarah's data and identify potential diagnoses. Reflective Capabilities: The system recognises its own limitations and provides transparency into its decision-making process. Feedback Loop: The system continuously learns from real-world cases, improving its diagnostic accuracy over time. Outcome: With the assistance of the reflective and self-aware diagnostic support system, healthcare professionals are able to consider a wider range of potential diagnoses and treatment options. The system's ability to provide explanations for its recommendations enhances trust and collaboration between the system and the medical team. In Sarah's case, the system identifies a rare autoimmune disorder that had been previously overlooked due to its atypical presentation. The early diagnosis allows for timely treatment, significantly improving Sarah's quality of life and preventing further health complications. Conclusion : This case study illustrates the critical need for reflective and self-aware systems in healthcare. By harnessing the power of data integration, machine learning, and a reflective approach, these systems can assist medical professionals in complex decision-making processes, reduce diagnostic errors, and ultimately enhance patient care and outcomes. In domains where human expertise is crucial but can be limited, such systems serve as valuable tools to augment human intelligence and improve decision quality. Core Reading

Read the article below to learn fundamental concepts, architectures, techniques, applications, and case studies in the exciting area of self-awareness in autonomous and computing systems. Self-Awareness for Autonomous Systems An outlook for self-awareness in computing systems Read the Introduction in Cognitive Networks: Towards Self-Aware Networks to get an understanding of how cognitive networks lead towards self-aware networks: Introduction Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [Self-Awareness for Autonomous Systems](#)
- [Link] [An outlook for self-awareness in computing systems](#)
- [Link] [Cognitive Networks: Towards Self-Aware Networks](#)
- [Link] [Introduction](#)

Autonomic Computing

Module 5 Autonomic Computing Icon Progress Bar (browser only) Focus This page introduces the concept of Autonomic Computing and discusses some related examples. Autonomic Computing Autonomic computing is inspired by analogy with the human body's nervous system. In the same way that the autonomic nervous system acts and reacts to stimuli to regulate and protect the human body, independent of the individual's conscious inputs, autonomic computing can support analogous behaviours in virtual computing systems. It is effectively self-managing. Autonomic systems are often described as “self-*” or “self-star” with the asterisk, or “star,” representing different properties of autonomic systems (self-organisation, self-maintenance). Autonomic systems are constructed as a group of locally interacting autonomous entities that cooperate to maintain system-wide functionality without any external control. Example: Smart Thermostat Imagine you have a smart thermostat installed in your home. This thermostat is an example of autonomic computing because it demonstrates several key characteristics of autonomic systems, such as self-configuration, self-optimisation, and self-healing. Self-Configuration: When you first install the thermostat, you provide it with some basic information, such as your desired temperature settings and your daily schedule. The thermostat uses this information to configure itself based on your preferences. It learns your temperature preferences and creates a schedule to optimise energy efficiency. Self-Optimisation: Over time, the thermostat starts to optimise its operations without your direct intervention. It continually monitors the indoor and outdoor temperatures, and it learns how long it takes to heat or cool your home effectively. As a result, it may adjust the heating or cooling cycle to be more energy-efficient while still maintaining your comfort. Self-Healing: If the smart thermostat detects a problem, like a malfunctioning sensor or a Wi-Fi connectivity issue, it can take corrective actions on its own. For example, if the Wi-Fi connection is lost, it can store and execute your pre-set temperature schedule until the connection is restored. This ensures that your home remains comfortable even during temporary disruptions. Self-Protection: The thermostat also has built-in safeguards to prevent overheating or overcooling. If it senses that the temperature is not changing as expected (possibly due to a malfunction), it can shut off the heating or cooling system to protect your home from extreme temperatures. Adaptability: As the seasons change, the thermostat adapts its behaviour without your intervention. For example, during the winter, it may increase the

heating cycle duration, while in the summer, it may increase the cooling cycle duration, aligning with the changing weather conditions. In this example, the smart thermostat acts autonomously to provide you with a comfortable and energy-efficient environment. It continually learns and adjusts to changing conditions, self-configures based on your preferences, optimizes its operations, and even takes corrective actions when needed. This is a simplified illustration of how autonomic computing systems work to make our lives easier and more efficient.

Key Terms See below for key terms you'll need to be familiar with when dealing with autonomic computing.

Autonomic components: Self-contained software or hardware modules with an embedded capacity for self-management, connected via input/outputs to other components in the system.

Bootstrapping: A self-starting process in a computer system, configured to automatically initiate other processes after the booting process has been initiated.

Multi-agent system: A system consisting of multiple separate agents, either software or hardware systems, that can cooperate and organise to solve problems.

Resource distribution: The locations of resources available to a computing system through various software or hardware components or networked computer systems.

Self-star properties: A list of component and system properties required for a computing system to be classified as an autonomic system.

Read More Autonomic Attributes and the Grand Challenge For an introduction to autonomic computing concepts and design, study the sections below in Autonomic Computing by Richard Murch.

Introduction Autonomic Computing Elements As computer systems have advanced from very basic technologies needing intense IT management toward autonomic systems that can self-manage, there have been four major stepping stones: self-optimizing, self-protecting, self-healing, and self-configuring. Each of these steps toward fully autonomic systems allows for more expansive computing while reducing the skill level required of the end users.

S ELF -S TAR S YSTEMS The concept of autonomic computing is based on autonomic systems found in nature. Examples of such systems include the autonomic nervous system of humans and the self-regulation of colonial insects such as bees and ants. In an autonomic system, the behaviors of individual components lead to higher-order self-maintenance properties of the group as a whole.

- self-stabilization, the ability to return to a stable state after a change in configuration;
- self-healing, the ability to recover from external damage or internal errors;
- self-organization, the ability to organize component parts and processes toward a goal;
- self-protection, the ability to combat external threats to operation; and
- self-optimization, the ability to manage all resources and components to optimize operation.

Autonomic systems may also display self-awareness and self-learning. Self-awareness in a computer system differs from self-awareness in a biological system. In a computer system, self-awareness is better defined as the system's knowledge of its internal components and configuration. Self-learning is the ability to learn from experiences without a user programming new information into the system.

https://science.jrank.org/computer-science/Autonomic_Computing.html nks to an external site. Source : https://www.researchgate.net/figure/Autonomic-Computing-Attributes-Autonomic-Computing-Attributes-Autonomic-Computing_fig3_324360420 External Reading Materials For an overview of the four aspects and for further detail on each one, read the following chapters in Autonomic Computing by Richard Murch.

Autonomic Computing Elements Self-Configuring Self-Optimising Self-Healing Self-Protecting Autonomic Products and Applications For examples of autonomic products and their applications, check out the section below in Autonomic Computing by Richard Murch.

Autonomic Products and Applications The Promise of Autonomic Systems The main goal of autonomic computing is to enable computer

systems to perform basic maintenance and optimization tasks on their own. Maximizing the number of automated functions that a computer can handle allows engineers and system administrators to focus on other activities. It also enhances ease of operation, especially for those less adept at data management or system maintenance. For instance, the bootstrapping system and the self-regulatory systems that detect and correct errors have made computing more friendly for the average user. Autonomic computer systems are particularly important in robotics, artificial intelligence (AI), and machine learning. These fields seek to design machines that can work unaided after initial setup and programming. The science of autonomic computing is still in its infancy, but it has been greatly enhanced by advancements in processing power and dynamic computer networking. For instance, the AI system Amelia, developed by former IT specialist Chetan Dube, not only responds to verbal queries and answers questions but can also learn by listening to human operators answer questions that it cannot answer. To some, systems that can learn and alter their own programming are the ultimate goal of autonomic design. Read More Activity Brainstorm one autonomic system idea from your daily life experience. Self-configuration: How will the system set itself up initially? What information will it need? Self-optimisation: How will the system continually optimise its performance? What variables will it monitor and adjust? Self-healing: How will the system detect and correct issues? What safeguards will be in place? Self-protection: How will the system protect itself and its users from potential risks? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [Read More](#)
- [Link] [Introduction](#)
- [Link] https://www.researchgate.net/figure/Autonomic-Computing-Attributes-Autonomic-Computing-Attributes-Autonomic-Computing_fig1.png
- [Link] [Autonomic Computing Elements](#)
- [Link] [Self-Configuring](#)
- [Link] [Self-Optimising](#)
- [Link] [Self-Healing](#)
- [Link] [Self-Protecting](#)
- [Link] [Autonomic Products and Applications](#)

Complex Systems and Artificial Life

Module 5 Complex Systems and Artificial Life Icon Progress Bar (browser only) Focus This page introduces the concept of Systems and discusses some related examples. Complex Systems A complex system is defined as a system whose properties are not fully explained by understanding its component parts. In computation, a complex system often represents a hard problem that cannot be solved within polynomial time. Complex ubiquitous systems consist of systems of many systems, such as systems of billions of networked elements structured into millions of interacting networks (complex networks), systems of many sensors or nanodevices (amorphous computing). Complex systems can arise when there are many possible combinations of interactions between different systems/sub-systems because of many interrelationships and interdependencies between them. Complex Ubiquitous System Example:

Smart City Components: Sensors and IoT Devices: Smart Cities deploy a wide range of sensors and Internet of Things (IoT) devices throughout the urban environment. These include environmental sensors, traffic cameras, waste management sensors, and more. Data Centers: Collected data is processed and stored in data centers, often in the cloud. These data centers analyze and manage vast amounts of information from the sensors. Communication Networks: High-speed communication networks, such as 5G, connect all the components, ensuring rapid data transmission and low latency. Citizen Engagement Platforms: Residents can interact with the city through mobile apps and websites, reporting issues, accessing services, and staying informed. Functionality: Traffic Management: Smart traffic lights and cameras optimize traffic flow, reducing congestion. Real-time data helps drivers find available parking spaces. Waste Management: Sensors in trash bins notify collection trucks when they are full, optimizing routes and reducing fuel consumption. Environmental Monitoring: Sensors track air quality, weather conditions, and pollution levels. The data can trigger alerts or city-wide actions to improve air quality. Energy Efficiency: Streetlights are equipped with motion sensors, turning on only when needed. Buildings use smart energy management systems to optimize power usage. Emergency Response: In case of natural disasters or emergencies, the system can provide real-time data to responders, helping them assess and manage the situation. User Experience: Residents can access real-time information on traffic, public transport, parking, and air quality through mobile apps. They can report issues like potholes or malfunctioning streetlights through the app, making the city more responsive to their needs. Environmental data can influence residents' daily choices, such as which routes to take for a morning jog or when to plan outdoor activities. Interconnectedness: All components are interconnected through a high-speed communication network. Data from various sensors and sources are analyzed in real-time. Personalization: The system can provide personalized recommendations to residents. For example, suggesting alternate routes during heavy traffic or informing residents about upcoming local events. Sustainability: By optimizing energy usage, traffic flow, and waste collection, Smart Cities reduce their environmental impact and contribute to sustainability goals. Benefits: Improved traffic flow and reduced congestion. Efficient waste management and reduced operational costs. Enhanced environmental monitoring and air quality improvements. More responsive and citizen-centric urban services. Lower energy consumption and reduced carbon footprint. A Smart City is a comprehensive example of a complex ubiquitous system that integrates various technologies and services to make urban living more efficient, sustainable, and user-friendly. It demonstrates how a city's infrastructure, services, and daily life can be transformed through advanced connectivity and data-driven decision-making.

Artificial Life

Artificial Life refers to systems that mimic natural life. They are characterised by: Having a finite lifetime from birth to death. Using selective reproduction. Offspring inheriting some of the traits of their parents. Using survival of the fittest (evolution). Supporting the ability to expand in numbers to command a space or habitat. Responding to stimuli in a habitat, acting to maintain it and adapting to it. Artificial Life systems are generally governed by biologically inspired algorithms and evolutionary computing. They involve computer algorithms that entail cumulative selections from a population of entities to solve a problem. The main types of evolutionary computing techniques include Genetic Algorithms (GA)/Genetic Programming (GP) and Evolutionary Strategies (ES)/Evolutionary Programming (EP). For a detailed explanation of Biologically Inspired Systems and Networks, study the chapters below from Cognitive Networks: Towards Self-Aware Networks by Qusay H. Mahmoud. You'll learn about different biological systems and their possible application in the field of computer networking .

Introduction Links to an external site. Chapter 1. Biologically Inspired Networking Swarm Intelligence Swarm Intelligence (SI) deals with natural and artificial systems composed of many individuals that coordinate using decentralized control and self-organization. In particular, SI focuses on the collective behaviors that result from the local interactions of the individuals with each other and with their environment. Examples of systems studied by SI include colonies of ants and termites, schools of fish, flocks of birds, herds of land animals, multi-robot systems, and certain computer programs designed to tackle optimization and data analysis problems. Swarm intelligence has been applied to various real-life projects and applications. Here are some examples: Ant Colony Optimization for Routing: Used to optimize routing in transportation and logistics by mimicking the foraging behavior of ants, leading to cost savings and reduced travel time. Swarm Robotics: Groups of small robots work together to perform tasks such as search and rescue operations, environmental monitoring, or agricultural tasks like planting and harvesting. Traffic Management: Optimizing traffic signal timings in urban areas based on real-time traffic data, reducing congestion, and improving traffic flow. Drone Swarms: Coordinated drones used for surveillance, agriculture, and entertainment, performing tasks such as synchronized light shows or data collection. Swarm-Based Search and Rescue: Used in disaster scenarios to search for survivors, covering larger areas and improving search efficiency. Optimizing Wireless Sensor Networks: In IoT applications, improving sensor network efficiency and prolonging battery life using swarm intelligence techniques. Resource Allocation in Cloud Computing: Optimizing resource allocation in cloud computing environments based on demand and usage patterns, improving performance, and reducing costs. Financial Market Forecasting: Using swarm intelligence algorithms to predict stock market trends based on collective trader behavior. Agricultural Applications: Applying swarm intelligence in precision agriculture, where drones and robots monitor crops, apply fertilizers, and perform tasks efficiently. Particle Swarm Optimization in Engineering: Applied in engineering design and optimization tasks such as structural design and aerodynamic profile optimization. These real-life projects demonstrate the versatility and effectiveness of swarm intelligence in solving complex problems across various domains. For more information on Swarm Intelligence, read the chapter below from Cognitive Networks: Towards Self-Aware Networks by Qusay H. Mahmoud. Swarm Intelligence For more information on evolutionary and adaptive systems and how they represent artificial life, read the chapter below from Cognitive Networks: Towards Self-Aware Networks by Qusay H. Mahmoud. Evolutionary and Adaptive Systems Resources For more information on the topics discussed, refer to the following resources: Introduction to Biologically Inspired Networking Chapter 1: Biologically Inspired Networking Swarm Intelligence Evolutionary and Adaptive Systems Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel Concluding Module 5 Well done, tau kē! You have reached the end of this module! What you achieved In this module, you considered concepts such as artificial life, autonomous and complex systems, reflective and self-aware systems. Coming up The next final module of the course will look at the Internet of Things (IoT).

Links & References

- [\[Link\] Introduction](#)
- [\[Link\] Chapter 1. Biologically Inspired Networking](#)
- [\[Link\] Swarm Intelligence](#)

- [Link] [Evolutionary and Adaptive Systems](#)

Module 6: Internet of Things

Module 6 Introduction

Module 6 Internet of Things: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, In this module, you will: Explore what the IoT is, why it matters, how it works and how you can build your own architecture Look into machine learning and why it is so crucial for IoT, including experimenting with Azure Machine Learning Studio to manage machine learning models Investigate IPv6 and how it gives longevity to IoT products, including 6LoWPAN Delve into CoAP and MQTT, two widely used data management protocols for the IoT Study Contiki, an open source operating system for the Internet of Things. Ngā Tikanga Paihere - Noa he tikanga or principle we will focus on in this module is Noa , which stands for benefit and opportunity. It's about balancing benefits and risks. Noa can be described as follows: Source: <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf>

Links & References

- [Link] <https://www.data.govt.nz/assets/data-ethics/Nga-Tikanga/Nga-Tikanga-Paihere-Guidelines-December-2020.pdf>

IoT Overview

Module 6 IoT Overview Icon Progress Bar (browser only) Focus This page introduces the concept of Internet of Things (IoT) and discusses some related examples. IoT Overview The Internet of Things is actually a pretty simple concept; it means taking all the physical places and things in the world and connecting them to the internet . Confusion arises not because the concept is so narrow and tightly defined, but rather because it is so broad and loosely defined. It can be hard to nail down the concept in your head when there are so many examples and possibilities in IoT. <https://www.leverage.com/iot-ebook/what-is-iot> Ultimately, IoT describes the network of physical objects, i.e., "things" embedded with sensors, software, and other technologies to connect and exchange data with other devices and systems over the Internet. Video: IoT Explained Why IoT Matters When something is connected to the internet, that means it can send information or receive information, or both. This ability to send and/or receive information makes things "smart." Let us use smartphones again as an example. Right now you can listen to just about any song in the world, but it is not because your phone actually has every song in the world stored on it. It is because every song in the world is stored somewhere else, but your phone can send information (asking for that song) and then receive information (streaming that song on your phone). To be smart, a thing does not need to have super storage or a supercomputer inside it - it just needs access to it. All a thing has to do is connect to super storage or to a supercomputer. On the Internet of Things, all the things that are being connected to the internet can be put into three categories: Things that collect information and then send it. Things that receive information and then act on it. Things that do both. All three of these have enormous benefits that compound on each other. IoT Functions 1. Collecting and Sending Information Sensors could be temperature sensors, motion sensors, moisture sensors, air quality sensors, light sensors, you name it. These sensors, along with a

connection, allow us to automatically collect information from the environment which, in turn, allows us to make more intelligent decisions. On a farm, automatically getting information about the soil moisture can tell farmers exactly when their crops need to be watered. Instead of watering too much (which can be an expensive over-use of irrigation systems) or watering too little (which can be an expensive loss of crops), the farmer can ensure that crops get exactly the right amount of water. This enables farmers to increase their crop yield while decreasing their associated expenses. Just as our sight, hearing, smell, touch, and taste allow us, humans, to make sense of the world, sensors allow machines (and the humans monitoring the machines) to make sense of the world.

2. Receiving and Acting on Information

We're all very familiar with machines getting information and then acting. Your printer receives a document and it prints it. Your car receives a signal from your car keys and the doors open. The examples are endless. Whether it's as simple as sending the command "turn on" or as complex as sending a 3D model to a 3D printer, we know that we can tell machines what to do from far away. So what? The real power of the Internet of Things arises when things can do both of the above. Things that collect information and send it, but also receive information and act on it.

3. Doing Both: The Goal of an IoT System

Let's quickly go back to the farming example. The sensors can collect information about the soil moisture to tell the farmer how much to water the crops, but you don't actually need the farmer. Instead, the irrigation system can automatically turn on as needed, based on how much moisture is in the soil. You can take it a step further too. If the irrigation system receives information about the weather from its internet connection, it can also know when it's going to rain and decide not to water the crops today because they'll be watered by the rain anyways. And it doesn't stop there! All this information about the soil moisture, how much the irrigation system is watering the crops, and how well the crops actually grow can be collected and sent to supercomputers that run amazing algorithms that can make sense of all this information. And that's just one kind of sensor. Add in other sensors like light, air quality, and temperature, and these algorithms can learn much, much more. With dozens, hundreds, thousands of farms all collecting this information, these algorithms can create incredible insights into how to make crops grow the best, helping to feed the world. And agriculture is just one of many applications of IoT.

Example: IoT Scenario in Precision Farming

IoT with Precision Farming

In a smart agriculture scenario, IoT devices are employed for precision farming. Various sensors, drones, and automated machinery are connected to a central system, providing farmers with real-time data and control over their operations.

- Soil Moisture Sensors:** These sensors can be embedded in the soil to measure moisture levels. The data is sent to the farmer's device, allowing them to optimize irrigation schedules, preventing over-watering or under-watering.
- Drones:** Drones equipped with cameras and sensors can survey large fields quickly. They capture images and data on crop health, pest infestations, and nutrient levels. This information is then analyzed to make informed decisions about crop management.
- Automated Tractors:** Tractors with IoT capabilities can be programmed to plow, sow, and harvest crops with precision. The farmer can control these machines remotely and receive updates on their progress.
- Weather Stations:** IoT-enabled weather stations on the farm provide real-time weather data. Farmers can receive alerts about upcoming storms or other adverse conditions, allowing them to take preventive measures.

Traditional Farming without IoT

In a traditional farming scenario without IoT, these processes would be more manual and less data-driven.

- Manual Soil Testing:** Farmers would need to manually collect soil samples and send them to a lab for analysis. The results would take time, and decisions would be based on less immediate and comprehensive information.
- Visual Crop Inspection:** Farmers would rely on

visual inspections of crops, which might not detect subtle signs of disease or stress as effectively as sensors and drones. **Manual Operation of Machinery:** Tractors and other machinery would require manual operation, making it less efficient and precise compared to automated, IoT-enabled machinery. **Limited Weather Information:** Without real-time weather data, farmers might not receive timely warnings about changing weather conditions. **Comparison Efficiency:** With IoT, precision farming is more efficient, as decisions are based on real-time data. Without IoT, farmers rely on more time-consuming and manual methods. **Resource Optimization:** IoT enables precise resource management, such as water and fertilizers, leading to cost savings and environmental sustainability. In a non-IoT scenario, resource usage may be less optimized. **Timely Decision-Making:** IoT allows farmers to make timely decisions based on immediate data. In a non-IoT scenario, there may be delays in obtaining and acting on critical information. In summary, IoT in agriculture enhances efficiency, precision, and decision-making, offering significant advantages over traditional farming methods. <https://www.leverage.com/iot-ebook/what-is-iot> Resources For a detailed understating of what IoT is, check out the following video. What is IoT Links to an external site. The Power of IoT: Examples and Applications Activity Choose three examples/applications of IoT you are most interested in and conduct further research on each of those. Then, capture your answers to the following questions: What area did you choose? What interests you about this area? How is the IoT intended to be used? How does the IoT enhance this area? Share your findings in the discussion forum. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://www.leverage.com/iot-ebook/what-is-iot>
- [Link] [What is IoT](#)
- [Link] [The Power of IoT: Examples and Applications](#)

How IoT Systems Work

Module 6 How IoT Systems Work Icon Progress Bar (browser only) Focus This page introduces the details of how IoT systems work and discusses some related examples. **How IoT Systems Work 1. Sensors/Devices** Sensors or devices collect data from their environment. This data could be as simple as a temperature reading or as complex as a full video feed. We use "sensors/devices" because multiple sensors can be bundled together or sensors can be part of a device that does more than just sense things. For example, your phone is a device that has multiple sensors (camera, accelerometer, GPS, etc.), but it can also perform many other actions. However, whether it's a standalone sensor or a full device, in this first step data is being collected from the environment by something . **2. Connectivity** Next, that data is sent to the cloud, but it needs a way to get there! The sensors/devices can be connected to the cloud through a variety of methods including cellular, satellite, WiFi, Bluetooth, low-power wide-area networks (LPWAN), connecting via a gateway/router or connecting directly to the internet via ethernet (don't worry, we'll explain more about what these all mean in our connectivity section). Each option has tradeoffs between power consumption, range, and bandwidth. Choosing which connectivity option is best comes down to the specific IoT application, but they all accomplish the same task: getting data to the cloud. **3. Data Processing** Once the data gets to the cloud (we'll cover what the cloud means in our data processing section)), software performs

some kind of processing on it. This could be very simple, such as checking that the temperature reading is within an acceptable range. Or it could also be very complex, such as using computer vision on video to identify objects (such as intruders on a property). But what happens when the temperature is too high or if there is an intruder on property? That's where the user comes in. 4. User Interface Next, the information is made useful to the end-user in some way. This could be via an alert to the user (email, text, notification, etc). For example, a text alert when the temperature is too high in the company's cold storage. A user might have an interface that allows them to proactively check in on the system. For example, a user might want to check the video feeds on various properties via a phone app or a web browser. However, it's not always a one-way street. Depending on the IoT application, the user may also be able to perform an action and affect the system. For example, the user might remotely adjust the temperature in the cold storage via an app on their phone. And some actions are performed automatically. Rather than waiting for you to adjust the temperature, the system could do it automatically via predefined rules. Rather than just call you to alert you of an intruder, the IoT system could also automatically notify security teams or relevant authorities. <https://www.leverage.com/iot-ebook/how-iot-systems-work> Check out the following videos on O'Reilly for a more in-depth explanation on IoT building blocks, connectivity and the internet. IoT Building Blocks Links to an external site. [9m 18s] IoT Connectivity Links to an external site. [6m 51s] IoT and Internet Links to an external site. [10m 01s] Example: IoT in Smart Agriculture Sensors and Devices In a smart agriculture IoT system, various sensors are deployed across the farm, collecting data about soil conditions, crop health, and environmental factors. Data Collection The sensors continuously collect data and send it to a central hub, which can be a cloud-based platform or an on-premises server. Connectivity IoT devices use wireless connectivity, such as Wi-Fi or cellular networks, to transmit data, ensuring real-time communication between the sensors and the central system. Cloud Computing The collected data is often processed and stored in the cloud, which provides the necessary infrastructure for handling large volumes of data and performing complex analyses. Data Analysis In the cloud, data analytics algorithms process the incoming information. For example, machine learning models can analyze images from drones to identify crop diseases or pests. Decision-Making The analysed data is translated into actionable insights, allowing farmers to make informed decisions based on real-time updates on soil moisture, crop conditions, and other factors. Automation and Control Based on the insights, automated systems can be triggered, such as an irrigation system that adjusts water usage or automated machinery that performs specific tasks. Automated machinery, such as tractors, can be remotely controlled and programmed for specific tasks. Alerts and Notifications The system can generate alerts and notifications, such as warnings about potential crop diseases, adverse weather conditions, or equipment malfunctions. User Interaction Farmers can interact with the system, providing manual inputs or adjustments based on their expertise and preferences. Feedback Loop The system continuously gathers feedback, including the outcomes of automated actions, user inputs, and the overall performance of the IoT devices. In summary, IoT systems in smart agriculture create an interconnected network of sensors, devices, and cloud-based analytics. This network enables real-time data collection, analysis, decision-making, and automation, ultimately enhancing the efficiency and productivity of agricultural operations. Reflective Activity Objective: Think critically about the impact of IoT on agriculture and its potential benefits and challenges. Personal Connection: Reflect on your understanding of traditional farming methods and how they compare to the concept of smart agriculture with IoT. Pros and Cons Analysis: List three potential advantages of implementing

IoT in agriculture and three challenges or concerns associated with it. Environmental Impact: Explore the environmental implications of smart agriculture. Future Vision: Envision the future of farming with IoT. Actionable Insights: Identify one actionable insight or idea that could enhance the integration of IoT in agriculture. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] <https://www.leverage.com/iot-ebook/how-iot-systems-work>
- [Link] [IoT Building Blocks](#)
- [Link] [IoT Connectivity](#)
- [Link] [IoT and Internet](#)

IoT Architecture

Module 6 IoT Architecture Icon Progress Bar (browser only) Focus This page introduces the requirements you need to start with to build your own IoT architecture, and what existing architectures you can work from, such as the ETSI M2M standardised architecture and the IoT architectural reference model. Additionally, some related examples are discussed. IoT Architecture Requirements IoT architecture is designed to facilitate the connection and interaction of various devices, sensors, and systems, enabling them to communicate and share data. The architecture of an IoT system typically involves several key components and follows specific models. First, we will talk about the IoT architecture requirements: Interoperability: Devices and systems from different manufacturers should be able to work together seamlessly. Scalability: IoT systems need to be scalable to accommodate a growing number of devices and data. Security & Reliability: IoT architecture must include robust security measures and reliability mechanisms due to the nature of IoT applications. Data Management: Efficient data management is an essential requirement. IoT Architecture Models Three-Tier IoT Architecture: It provides clear separation of concerns through three layers: the perception layer, the network layer, and the application layer. Service-Oriented Architecture: This model breaks down the architecture into modular services that communicate with each other. Event-driven Architecture: This model focuses on changes such as device status, where any change triggers a specific action. Understanding these requirements and models is essential for designing effective and efficient IoT systems tailored to specific use cases and industries. Intelligent Lighting Example Let us see how our IoT architecture elements work together by examining the example of smart yard lighting as a part of a smart home – a bright illustration of how an IoT solution simultaneously contributes to user convenience and energy efficiency. Basic components Sensors take data from the environment (e.g., daylight, sounds, people's movements). Lamps are equipped with actuators to switch the light on and off. A data lake stores raw data coming from sensors. A big data warehouse contains the extracted information about smart home dwellers' behaviour, energy costs, and more. Manual monitoring and control Users control the smart lighting system with a mobile app featuring a map of the yard. The app allows users to see which lights are on and off and send commands to the control applications that further transmit them to lamp actuators. Data analytics Analyzing the way users apply smart lighting, their schedules (either provided by users or identified by the smart system), and other info gathered with sensors, data analysts can make and update the algorithms for control

applications. Data analytics also helps in assessing the effectiveness of the IoT system and revealing problems in the way the system works. For example, if a user switches off the light right after a system automatically switches it on and vice versa, there might be gaps in the algorithms, and it's necessary to address them as soon as possible. Automatic control options and pitfalls The sensors monitoring natural light send the data about the light to the cloud. When the daylight is not enough (according to the previously stated threshold), the control apps send automatic commands to the actuators to switch on the lamps. The rest of the time the lamps are switched off. However, a lighting system can be "baffled" by street illumination, lamps from neighboring yards and any other sources. Extraneous light captured by sensors can make the smart system conclude that it's enough light, and lighting should be switched off. Thus, it makes sense to give the smart system a better understanding of the factors that influence lighting and accumulate these data in the cloud. When sensors monitor motions and sounds, it's not enough just to switch on the light when movements or sounds are identified in the yard or switch all the lamps off in the silence. Movements and sounds can be produced, for example, by pets, and cloud applications should distinguish between human voices and movements and those of pets. The same is about the noises coming from the street and neighboring houses and other sounds. To address this issue, it's possible to store the examples of various sounds in the cloud and compare them with the sounds coming from the sensors. Machine learning Intelligent lighting can apply models generated by machine learning, for example, to recognize the patterns of smart home owners' behavior (leaving home at 8 am, coming back at 7 pm) and accordingly adjust the time when lights are switched on and off (for example, switch the lamps on 5 minutes before they will be needed). Analyzing users' behavior in a long-time perspective, a smart system can develop advanced behavior. For example, when sensors don't identify typical movements and voices of home inhabitants, a smart system can "suppose" that smart home dwellers are on a holiday and adjust the behavior: for example, occasionally switch on the lights to give the impression that the house is not empty (for security reasons), but do not keep the lights on all the time to reduce energy consumption. User management options To ensure efficient user management, the smart lighting system can be designed for several users with role distribution: for example, owner, inhabitants, guests. In this case, the user with the title "owner" will have full control over the system (including changing the patterns of smart light behavior and monitoring the status of the yard lamps) and priorities in giving commands (when several users give contradicting commands, an owner's command will be the one control apps execute), while other users will have access to a limited number of the system's functions. "Inhabitants" will be enabled to switch on and off the lamps with no opportunity to change settings. "Guests" will be able to switch on and off the light in some parts of the house and have no access to controlling the lights, for example, near the garage. Apart from role distribution, it's essential to consider ownership (as soon as one system can control over 100 thousand households, and it's important that a dweller of a smart home manages the lighting in his yard, and not the one of a neighbor).

<https://www.scnsoft.com/blog/iot-architecture-in-a-nutshell-and-how-it-works> Watch and Learn Watch the video tutorials below to learn more about IoT architecture requirements and models. IoT Architecture Requirements [6m 23s] Architecture Models Activity Consider how IoT can contribute to the development of smart grids in rural areas, optimising energy consumption on farms and promoting sustainability. Identify the components of IoT architecture. Consider how the components can be integrated with the existing infrastructure. Consider the scalability and flexibility for future growth. Need Help? Looking for help? Seeking

an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [info gathered with sensors](#)
- [Link] <https://www.scnsoft.com/blog/iot-architecture-in-a-nutshell-and-how-it-works>
- [Link] [IoT Architecture Requirements](#)
- [Link] [Architecture Models](#)

IoT and Machine Learning

Module 6 IoT and Machine Learning Icon Progress Bar (browser only) Focus This page focuses on IoT and Machine Learning and discusses some related examples. Example: Smart Agriculture with IoT and Machine Learning Business Name: KiwiFarmTech Overview: KiwiFarmTech is an innovative agribusiness in New Zealand that specializes in integrating IoT and Machine Learning technologies to enhance agricultural practices. Application of IoT: KiwiFarmTech employs a network of IoT devices across their farms. These devices include soil sensors, weather stations, and livestock trackers. Soil sensors continuously monitor moisture levels and nutrient content, providing real-time data to farmers. Weather stations collect atmospheric data, and livestock trackers monitor the health and location of animals. Integration of Machine Learning: The collected data from IoT devices is fed into a centralized platform that utilizes Machine Learning algorithms. Predictive Analytics: Machine Learning models analyze historical weather and soil data to predict optimal planting times, helping farmers make informed decisions about crop rotations and irrigation scheduling. Livestock Health Monitoring: ML algorithms analyze patterns in livestock data to predict health issues. For example, changes in movement patterns might indicate illness, prompting early intervention. Benefits: Precision Farming: Farmers can precisely optimize resource usage such as water and fertilizers based on real-time data, leading to increased efficiency and reduced environmental impact. Maximized Yields: Predictive analytics and data-driven decision-making contribute to higher crop yields and healthier livestock. Cost Reduction: Efficient resource use and early detection of potential issues reduce operational costs. User Experience: Farmers can access a user-friendly dashboard on their smartphones or computers. This dashboard provides a comprehensive overview of the farm's status, including soil conditions, weather forecasts, and livestock health. Automated alerts notify farmers of critical events, enabling proactive responses. Sustainability: KiwiFarmTech emphasizes sustainable farming practices. By optimizing resource use, the business contributes to environmental conservation, aligning with New Zealand's commitment to eco-friendly agriculture. This example illustrates how a New Zealand-based business can leverage IoT and Machine Learning to revolutionize traditional industries, enhancing productivity, sustainability, and overall efficiency. Watch the video below for an introduction to machine learning and related concepts. An Introduction to Machine Learning Links to an external site. Microsoft Azure Machine Learning Studio Azure Machine Learning Studio is a GUI-based integrated development environment for constructing and operationalising machine learning workflow on Azure. It provides a quick drag-and-drop workflow that simplifies and accelerates the process of building, testing, and deploying machine learning models for the entire data science team, from beginners to professionals. Spend some time exploring the site below to become familiar with the Microsoft Azure

Machine Learning Studio and its documentation. Azure Machine Learning documentation Activity Reflect on your understanding and share your reflections with other students in the discussion section. You can also attempt the activity below. Spend some time reading and following the prompts in chapters 03, 05 and 06 in the guide below. Microsoft Machine Learning - Microsoft Azure Essentials Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [An Introduction to Machine Learning](#)
- [Link] [Azure Machine Learning documentation](#)
- [Link] [Microsoft Machine Learning - Microsoft Azure Essentials](#)

IPv6 and IoT

Module 6 IPv6 and IoT Icon Progress Bar (browser only) Focus This page introduces IPv6 and IoT concepts and discusses their applications and challenges with real-life scenarios. Additionally, some external links to short videos and reading material are provided.

Introduction to IPv6 IPv6 is the next-generation Internet Protocol (IP) address standard intended to supplement and eventually replace IPv4, the protocol many Internet services still use today. Every device connected to the Internet requires a numerical IP address to communicate with other devices. The original IPv4 address scheme is running out of addresses due to the proliferation of so many connected devices. The primary function of IPv6 is to allow for more unique TCP/IP address identifiers to be created, now that we have exhausted the 4.3 billion addresses created with IPv4. This is one of the main reasons why IPv6 is such an important innovation for the Internet of Things (IoT). While IPv4 addresses could not meet the demand for IoT products, IPv6 provides a platform for IoT products to operate on for a long time.

Three Reasons Why IPv6 Is Important For the Internet Of Things

1. Security With billions of new smart products being created every day, security is an important thought in the back of all IoT engineers' minds. Organizations and individuals have learned of the real and imminent threat that hackers pose in past years, but the IoT brings up a whole new line of security intricacies. Hacking a secure network and harvesting millions of credit card numbers is terrible—but if someone with ill intentions was to hack into a smart city, or a neighborhood of smart houses, the outcome could be far more catastrophic. You can tell why IoT security is very important—and the good news is that IPv6 offers better security solutions than its predecessor, largely due to IPSec. For one thing, IPv6 can run end-to-end encryption. While this technology was retrofitted into IPv4, it remains an extra option that is not universally used. The encryption and integrity-checking used in current virtual private networks (VPNs) are a standard component in IPv6, available for all connections and supported by all compatible devices and systems. Widespread adoption of IPv6 will therefore make “man-in-the-middle” attacks—i.e., thinking that you’re signing into a secure bank log in when you’re actually walking into a cyber “trap”—significantly more difficult. IPv6 also supports more-secure name resolution. The Secure Neighbor Discovery (SEND) protocol is capable of enabling cryptographic confirmation that a host is who it claims to be at the time of the connection. This renders Address Resolution Protocol (ARP) poisoning and other naming-based attacks more difficult. And while IPv6 isn’t a replacement for application- or service-layer verification, it still offers an improved level of

trust in connections. With IPv4, it's fairly easy for an attacker to redirect traffic between two legitimate hosts and manipulate the conversation or at least observe it—but IPv6 makes this very difficult. These added security features depend entirely on proper design and implementation of IPv6, and the more complex, flexible infrastructure of IPv6 makes this process more difficult. Nevertheless, if properly configured, IPv6 networking will be significantly more secure than IPv4 by a longshot. 2. Scalability According to a report put out by Gartner, 25 billion “things” will be connected to the internet by the year 2020. That's a pretty incredible estimation, considering the same report notes that 4.9 billion devices will be connected in 2015. This purported 400% increase in growth in only five years sheds some light on how much exponential IoT growth we can expect to see in the next 10, 20, or even 50 years. Given these numbers, it's easy to understand why IPv6 (and its trillions upon trillions of new addresses) are important for IoT devices. Creators of IoT products that are connected over TCP/IP can rest assured that there will be a unique identifier available for their devices for a long, long time. 3. Connectability With billions of new IoT devices entering the market each year, connectability—i.e., allowing network-connected devices to “speak” to each other—is vital. With IPv4, there were quite a few issues with allowing IoT products to speak with one another. Network Address Translation (NAT) posed one of these major issues. NAT was created as a workaround for organizations who needed multiple people and devices to be able to work off of the same IPv4 address. Not only does this pose a security issue (which we'll talk about in a moment), but it also poses a difficult issue for IoT products. IPv6 allows IoT products to be uniquely addressable without having to work around all of the traditional NAT and firewall issues. Larger and more advanced host devices have all sorts of tools to make working with firewalls and NAT routers easier, but small IoT endpoints do not. By using IPv6, many of these issues become easier for TCP/IP enabled IoT devices to handle.

<https://www.link-labs.com/blog/why-ipv6-is-important-for-internet-of-things> Additional Resources Read/watch the resources below to find out more about IPv6 in relation to IoT. IPv4 vs. IPv6 Driving Forces for IPv6 in IoT IPv6 and its Place in an IoT Architecture Example: Smart Retail in New Zealand Imagine a retail chain in New Zealand that embraces smart technologies for an enhanced customer experience. Smart Store Infrastructure: IoT devices like smart shelves, RFID tags, and customer tracking sensors, all IPv6-enabled, ensure seamless communication and data transfer. IPv6 Benefits: The extensive address space accommodates the large number of IoT devices, each with a unique and globally accessible IPv6 address. Data-Driven Shopping Experience: Real-time inventory data and customer tracking help optimize stock levels and customer engagement. Personalized Shopping: The system analyzes customer preferences and sends personalized offers to their smartphones. Efficient Operations: IoT devices manage energy and inventory efficiently, improving operational effectiveness. Benefits: Enhanced customer experience with personalized services. Improved operational efficiency through real-time data analytics. Sustainable practices with energy-efficient operations. This example demonstrates how IPv6 and IoT can transform traditional retail into a smart, data-driven environment, providing benefits for both customers and businesses. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [security is an important thought in the back of all IoT engineers' minds](#)
- [Link] [smart houses](#)

- [\[Link\] IoT security](#)
- [\[Link\] a report put out by Gartner](#)
- [\[Link\] https://www.link-labs.com/blog/why-ipv6-is-important-for-internet-of-things](https://www.link-labs.com/blog/why-ipv6-is-important-for-internet-of-things)
- [\[Link\] IPv4 vs. IPv6](#)
- [\[Link\] Driving Forces for IPv6 in IoT](#)
- [\[Link\] IPv6 and its Place in an IoT Architecture](#)

6LoWPAN and IoT

Module 6 6LoWPAN and IoT Systems Icon Progress Bar (browser only) Focus This page introduces 6LoWPAN and IoT systems and discusses the applications and challenges. Some external links to short videos and reading material are also provided. 6LoWPAN Overview 6LoWPAN refers to IPv6 over Low power Wireless Personal Area Networks. It is a somewhat contorted acronym that combines the latest version of the internet protocol, i.e. IPv6, and Low-power Wireless Personal Area Networks (LoWPAN). 6LoWPAN allows for the smallest devices with limited processing ability to transmit information wirelessly using an internet protocol. This low power wireless mesh network, where every node has its own IPv6 address, allows the node to connect directly with the Internet using open standards. Advantages of 6LoWPAN It works great with open IP standard including TCP, UDP, HTTP, COAP, MATT and web-sockets. It offers end-to-end IP addressable nodes. There's no need for a gateway, only a router which can connect the 6LoWPAN network to IP. It supports self-healing, robust and scalable mesh routing. Offers one-to-many & many-to-one routing. The 6LoWPAN mesh routers can route data to other nodes in the network. In a 6LoWPAN network, leaf nodes can sleep for a long duration of time. It also offers thorough support for the PHY layer which gives freedom of frequency band & physical layer, which can be used across multiple communication platforms like Ethernet, Wi-Fi, 802.15.4 or Sub-1GHz ISM with interoperability at the IP level. It is a standard: RFC6282 6LoWPAN Application Areas With many low power wireless sensor networks and other forms of wireless networks designed to tackle specific problems, it is essential that any new wireless system has a defined area which it addresses. While there are many forms of wireless networks including wireless sensor networks, 6LoWPAN addresses an area that is currently not addressed by any other system, for example, that of using IP, and in particular IPv6 to carry the data. The overall system is aimed at providing wireless internet connectivity at low data rates and with a low duty cycle. However, there are many applications where 6LoWPAN is being used: Automation: 6LoWPAN offers many opportunities for automation in different fields. Industrial monitoring: Industrial plants and automated factories can use 6LoWPAN to monitor data and connect to the cloud for analysis. Smart Grid: Smart grids use IPv6 to connect smart meters and other devices in a micro mesh network, sending data back to the grid operator's system. Smart Home: IoT devices in homes can be connected using IPv6 for better efficiency and functionality. 6LoWPAN Security 6LoWPAN can use AES-128 link layer security which is defined in IEEE 802.15.4. This provides link authentication and encryption. Further security is provided by the transport layer security mechanisms. This is defined in RFC 5246 and runs over TCP. For systems where UDP is used, the transport layer protocol defined under RFC 6347 can be used. Good links for further reading: https://www.eetimes.com/document.asp?doc_id=1278794# <https://tools.ietf.org/html/rfc6282> Radiocrafts' RF Modules Supporting 6LoWPAN Radiocrafts' RC1882CEF-IPM RIIM Module supports 6LoWPAN. RIIM™ (Radiocrafts Industrial IP Mesh) is an embedded RF system

designed for easy use with direct IP addressing. It uses the IEEE802.15.4 g/e standard and includes Intelligent C-programmable I/O (ICI), supporting Mist Computing. RIIM™ requires no license or subscription fee. Read more about RIIM here

<https://radiocrafts.com/technologies/6lowpan/> Core Reading For more information on 6LoWPAN, study Chapter 2 (Introduction to 6LoWPAN) of the book below. IoT in Five Days Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] https://www.eetimes.com/document.asp?doc_id=1278794#
- [Link] <https://tools.ietf.org/html/rfc6282>
- [Link] [Read more about RIIM here](#)
- [Link] <https://radiocrafts.com/technologies/6lowpan/>

CoAP, MQTT and IoT

Module 6 CoAP, MQTT, and IoT Icon Progress Bar (browser only) Focus This page introduces CoAP and MQTT protocols, discussing their applications and challenges. It includes external links to short videos and reading materials. Overview of CoAP and MQTT CoAP (Constrained Application Protocol) and MQTT (Message Queuing Telemetry Transport) are both lightweight and efficient protocols designed for communication in the Internet of Things (IoT). Each protocol serves different purposes, and their selection depends on specific use cases and requirements. CoAP Overview Purpose: CoAP is designed for resource-constrained devices and low-power, low-bandwidth networks. Communication Model: It follows a simple request-response model similar to HTTP, making it suitable for RESTful interactions in constrained environments. Transport Protocol: CoAP typically uses UDP (User Datagram Protocol) as the underlying transport layer. Scalability: Well-suited for large-scale IoT deployments due to its lightweight nature. CoAP Use Cases Commonly used in scenarios where devices need to communicate with each other and with web servers in constrained environments (e.g., smart homes, industrial IoT). CoAP Advantages Efficiency: CoAP minimizes protocol overhead, making it efficient for constrained devices. RESTful Design: Supports RESTful principles, allowing for easy integration with web services. CoAP Challenges Limited to simple request-response interactions, which might be insufficient for more complex IoT scenarios. MQTT Overview Purpose: MQTT is designed for efficient communication between devices, especially in scenarios where low bandwidth and high latency are common. Communication Model: Follows a publish-subscribe model, enabling asynchronous communication between devices. Transport Protocol: MQTT can use TCP/IP or WebSocket as transport protocols, providing flexibility in network configurations. Scalability: Well-suited for scenarios with a large number of devices and real-time data requirements. MQTT Use Cases Widely used in applications where real-time data updates, event-driven architectures, and telemetry data are crucial (e.g., industrial automation, smart cities). MQTT Advantages Asynchronous Communication: Supports asynchronous communication, allowing devices to publish data when ready, and subscribers to receive updates in real-time. Reliability: Provides mechanisms for reliable message delivery, even in unreliable network conditions. MQTT Challenges Increased protocol overhead compared to CoAP, which might be a concern for highly constrained devices. Core Reading To find out more about CoAP, explore the resources below: Understanding CoAP CoAP

To find out more about MQTT, spend some time on the resources below: MQTT Understanding MQTT Further Reading For more information on CoAP and MQTT, study Chapter 5 (CoAP, MQTT and HTTP) of the book below. IoT in Five Days Integration in IoT Complementary Use: CoAP and MQTT can be used together in an IoT ecosystem. CoAP can be used for device-to-device communication, while MQTT can handle more complex scenarios involving data aggregation, analytics, and integration with cloud services. Scalability and Resource Considerations: The choice between CoAP and MQTT depends on factors such as the nature of the devices, network constraints, scalability requirements, and the overall IoT architecture. In summary, CoAP and MQTT serve different purposes in the IoT landscape, and their selection depends on the specific needs and constraints of the application. Integrating both protocols can provide a comprehensive solution for diverse IoT scenarios. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel

Links & References

- [Link] [Understanding CoAP](#)
- [Link] [CoAP](#)
- [Link] [MQTT](#)
- [Link] [Understanding MQTT](#)
- [Link] [IoT in Five Days](#)

Understanding Contiki

Module 6 Understanding Contiki Icon Progress Bar (browser only) Focus This page introduces the Contiki operating system. Real-life scenarios along with external links to short videos and reading material are also provided. Overview of Contiki Contiki is an open-source operating system for the Internet of Things. Contiki connects tiny low-cost, low-power microcontrollers to the Internet. Contiki is a powerful toolbox for building complex wireless systems. Contiki provides powerful low-power Internet communication. It supports fully standard IPv6 and IPv4, along with recent low-power wireless standards: 6lowpan, RPL, CoAP. With Contiki's ContikiMAC and sleepy routers, even wireless routers can be battery-operated. Contiki provides multitasking and a built-in Internet Protocol Suite (TCP/IP stack), yet needs only about 10 kilobytes of RAM and 30 kilobytes of ROM. A full system, including a graphical user interface, needs about 30 kilobytes of RAM. Contiki applications are written in standard C, and with the Cooja simulator, Contiki networks can be emulated before being burned into hardware. Instant Contiki provides an entire development environment in a single download. Contiki is open-source, which means that the source is and always will be available. It may be used in both commercial and non-commercial systems without restrictions. <https://iotbyhvm.ooo/contiki-the-open-source-os-for-iot/> Extant uses for Contiki include systems for street lighting, sound monitoring for smart cities, radiation monitoring, and alarms. Scenario: Smart Agriculture with Contiki in New Zealand Background: New Zealand has a significant agricultural sector, and there is a growing interest in leveraging IoT technologies to enhance farming practices. Implementation: Farmers in rural New Zealand deploy IoT sensors and devices across their fields to monitor various parameters, such as soil moisture, temperature, and crop health. Contiki OS is chosen as the operating system for these IoT devices due to its lightweight nature and energy-efficient features. Components: Soil Moisture Sensors: Contiki-powered IoT devices

equipped with soil moisture sensors are buried in different sections of the field. Weather Stations: Weather stations with Contiki OS collect data on temperature, humidity, and sunlight exposure. Gateway Device: A central gateway device, also running Contiki, is strategically placed to collect data from the distributed sensors. Data Analysis Server: The collected data is transmitted to a centralized server for analysis. This server employs Contiki for efficient data processing. Functionality: Real-time Monitoring: Farmers can monitor real-time data through a web-based dashboard, showing soil conditions and weather patterns across the farm. Automated Irrigation: Contiki OS facilitates communication between soil moisture sensors and irrigation systems. Automated irrigation is triggered based on real-time soil moisture levels. Weather-Adaptive Farming: The system, powered by Contiki, adjusts farming practices based on weather predictions and historical data, optimizing crop yield. Energy Efficiency: Contiki's low-power capabilities ensure that IoT devices operate efficiently, conserving energy and allowing for extended battery life. Benefits: Resource Optimization: Farmers can optimize resource usage, such as water and energy, leading to cost savings. Increased Yield: Smart agriculture practices, enabled by Contiki, contribute to increased crop yield. Environmental Sustainability: Efficient resource usage and reduced waste contribute to environmentally sustainable farming. Practice For more information on Contiki, study Chapter 3 (Introduction to Contiki) of the book below and follow the prompts to start using Contiki. This chapter will help you to practice Contiki by installing, testing, and implementing it on real hardware devices. Internet of Things in Five Days Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8106 Support and Feedback Channel Concluding Module 6 Well done, tau kē! You have reached the end of this module! What you achieved In this module, you learned what the Internet of Things is and how it is implemented. Coming up This was the last module of this course. Well done on reaching the end! We hope you enjoyed your learning journey!

Links & References

- [Link] <https://iotbyhvm.ooo/contiki-the-open-source-os-for-iot/>

Assessments / Assignments

Assessment 1: Research and Requirements

Due: 2026-09-04T11:59:59Z | Points: 100

IT8106 Assessment 1: Research and Requirements This assessment contributes to 30% of the final course grade Assessment Overview This assessment asks you to model the key ubiquitous computing properties of a system in any given scenario. You will utilise and evaluate different interaction design models to meet the real-world requirements along with the skills of designing a blueprint for the development of actual systems. Research and requirements assessment will be analysed, functional and non-functional requirements will be performed and may include physical, virtual ICT or human environment. You will also gain specialised knowledge of models and techniques used in ubiquitous computing and intelligent systems and demonstrate analytical skills for experiments and solutions designs of real work problems. Learning Outcomes This assessment contributes to: LO 1: Apply key ubiquitous computing properties to the architectural design of ubiquitous computing systems. LO 2: Apply service architecture models of sensor technology and sensor networks to model context aware and

adaptive systems. LO 3: Evaluate the efficacy of interaction designs from ubiquitous computing systems. Turnitin Canvas Plagiarism Framework This assignment will be checked for plagiarism and originality using the Turnitin Canvas Plagiarism Framework. You can learn more about this system by reading What is the Turnitin Canvas Plagiarism Framework . Before submitting your final version, you have the chance to upload drafts to Turnitin Draft Coach. To do this, read the sections What is Turnitin Draft Coach and How to use Turnitin Draft Coach and follow the instructions provided there for drafting your assignment. The similarity threshold is 15%. Conditions You can access all course materials and any other resources you wish to use as you work on this assessment. The work you submit must be your own work. You can ask your lecturer to clarify the instructions and for advice, but they cannot assist you in completing the tasks required – you must carry out the tasks yourself! This assessment must be submitted in Week 6 . Your lecturer will confirm the due date. Scenario: Smart Utility System SmartBuild has been contracted by Kāinga Ora (the New Zealand Government organisation responsible for social housing) to develop a Smart Utility System. This system will be used in all social housing new builds throughout the Wellington Region. SmartBuild is a business that provides solutions to individuals as well as organisations to utilise their energy more efficiently by introducing a ubiquitous system for the whole buildings. The business was established in 2017 and has experienced rapid growth over the past two years. Here are some examples of the projects they have been engaged in: Smart metres for gas and electricity, these generate real time resource utilization data that offers comprehensive insight to the users enabling them to manage the usage more efficiently. Some examples are: Customers can save their expenses based upon the dynamic rates, timings of usage and energy company's policies. It allows for better and more efficient responses and decisions in times of high demand and peak pricing. Customers can also be provided with a better 'Direct Load Control' which is a form of demand response. Practically, this means more energy consuming devices can be controlled remotely at the time of peak demand. Different types of energy sources can be selected and opted for, based upon the usage and availability, e.g., solar, wind, hydro powered etc. Autonomous systems and products that enable the systems' users to manage and configure them to use remotely. Technologies implemented in 'Smart Homes, Smart Hospitals, Smart Cars, Smart Buildings and Smart Cities' etc. can help in managing energy efficiently. Context-aware Systems that can detect and react according to the environment and physical contexts. Examples are context-aware devices that can react and turn themselves on or off based upon the context. These systems have varied based on the client requirements. Decisions such as will the user have any control over reconfiguring or will that be all pre-set have resulted in different systems being design. Some specific examples are: Context aware activity and behaviour monitoring systems: These systems actively monitor the activities that the user is performing or any change in their behavioural patterns that links to the usage of different devices inside a house e.g., a cooling and heating system that activates itself based upon the user's activities and context. Context-aware systems that work collaboratively towards the overall goal of efficient energy utilisation and can decide for example which lights to turn on and which to remain off when someone is nearby. Applications like this have often been part of a "Smart Grid" technology. Your manager has asked you to prepare a report on possible Smart Utility Systems for the Kāinga Ora (Housing NZ) project. This will be used by the Senior Management Team (SMT) to decide on and develop the project plan for the Kāinga Ora Project. Tasks Task 1: User Case Scenarios Your job is to put together at least three use case scenarios for the Smart Utility System. Each use case will provide a narrative that describes the user's requirements and evaluates their

experience of the functionality of the system. When considering the use case scenarios, analyse both the functional and non-functional requirements of the Smart Utility System. These may include the Physical, Virtual ICT or Human Environment. Include diagrams or flow charts to further detail your use case scenarios. Each scenario should be around 400 words, exclusive of diagrams or flow charts.

Task 2: The Characteristics of your Ubiquitous Computer (UbiCom) System Considering the core properties of a UbiCom system and your use case scenarios, describe what the characteristics of your utility regulation system proposal will be. Your proposal will need to consider the diversity of the real-world environment. The real-life environment is social housing in New Zealand. Thinking through the real-life applications of your use case scenarios, discuss and justify the reasons for your proposal in relation to key smart utility system design. Make sure you consider all three use cases as a part of a unified system and justify what core UbiCom properties should be part of this unified system. Discuss whether the unified system has to fully support all of the five ubiquitous system properties. This section of your report should be around 1200 words, exclusive of diagrams or flow charts.

Task 3: Research solutions and technologies for utility regulation In Tasks 1 and 2 you have explored possible user case scenarios and made initial decisions on the core properties of your UbiCom system. In this task you need to take your proposal and explore what solutions and technologies are available to deliver your proposed solution requirements. Research and evaluate existing solutions and technologies for utility regulation. When evaluating the different sensing technologies or/and devices to deliver your proposed solution, make sure you justify your suggestions. Diagrams can be used to support your justification. Compare and contrast the efficacy of the solutions, including an analysis of their strengths and limitations. You can do this in the context of the proposal you outlined in Task 2. This section of your report should be around 1200 words, exclusive of diagrams or flow charts.

Task 4: Recommendations Your report will be used by the SMT to develop the project plan for the Kāinga Ora (Housing NZ) project. In your report conclusion, clearly summarise your proposed smart utility system, including your recommended solutions and technology based on your research findings.

Submission Instructions When you have completed Tasks 1 to 4, upload all of the assessment documents to Canvas in PDF, DOCX, DOC or ODT format.

Policies **Success Criteria** To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed. You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be awarded 50%.

Extensions and Late Submissions An extension can only be granted under exceptional conditions normally for a maximum of (5 working days) , unless there is evidence to support a longer timeframe. Complete and return the following form to your Course Lecturer: [Assessment Extension Request Form.pdf](#)

Assessments received one (1) to five (5) working days past the due date, or an approved extended due date, will be marked but will receive a mark capped at the pass mark i.e. 50% for graded assessments. Assessments received more than five (5) working days past the due date, or approved extended due date, will not be marked and will receive a mark of zero or fail.

Academic Integrity and Academic Misconduct Policy Whitecliffe is committed to ensuring the highest level of academic integrity and expect staff and students to conduct themselves as responsible and honest members of the academic community. Where there is evidence that breaches of academic integrity may have occurred, the course of Academic Integrity and Academic Misconduct Policy will be followed. This policy

considers academic integrity to mean honesty in all matters relating to academic endeavours. Breaches in academic integrity, or academic misconduct, include all forms of cheating incorporates plagiarism (intentional or unintentional), unauthorised collaboration, impersonation, falsification and collusion. Declaration By submitting your work, you are indicating that you agree to the following declaration: "The work presented in this assessment is to the best of my knowledge original, except as acknowledged in the text, and the material has not already been submitted, either in whole or in part, for any academic award at this or any other tertiary institution. I promise not to share this assessment in part or whole with any other student at Whitecliffe or outside this campus."

Links & References

- [Link] [Assessment Extension Request Form.pdf](#)
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Assessment 2: Evaluative Design Solution Report

Due: 2026-11-06T10:59:59Z | Points: 100

IT8016 Assessment 2: Evaluative Design Solution Report This assessment contributes to 70% of the final course grade Assessment Overview In this assessment, you'll work on designing the solution to a real-world problem from a Ubiquitous Computing perspective. You will utilise and evaluate different interaction design models to meet real-world requirements along with designing a blueprint for the development of actual systems. You will encounter design issues, design constraints and choices, and consider how a part of your solution can be implemented using a mobile device software API. Learning Outcomes This assessment contributes to: LO 2: Apply service architecture models of sensor technology and sensor networks to model context aware and adaptive systems. LO 3: Evaluate the efficacy of interaction designs from ubiquitous computing systems. LO 4: Justify the design of autonomous systems to support self-operation in dynamic and diverse environments. Conditions You can access all course materials and any other resources you wish to use as you work on this assessment. The work you submit must be your own work. You can ask your lecturer to clarify the instructions and for advice, but they cannot assist you in completing the tasks required – you must carry out the tasks yourself! Scenario This assessment follows on from Assessment 1. Consider the scenario that you have worked on in Assessment 1. Your proposed Smart Utility System design has been accepted by Kāinga Ora and now your company, Smart Build, has been contracted to move to the next stage and submit design options and plans. Your manager has asked you to complete a report covering the following five tasks as the foundation for the design option discussions. Tasks Task 1: System Design Design an overall system model for the IoT-based Smart Utility monitoring system based on the functionalities and requirements mentioned in Assessment 1. Your proposed model must apply the core properties and key features of ubiquitous computing, such as human-computer interactions, context-awareness, adaptiveness, intelligence, autonomic computing and mobile computing to access/record data on the move. You must include these when designing the smart devices, services and sensor technology required for implementing your proposed system. Include in discussion of your design the application of any AI/agent techniques that could be used and justify your motivation for including them in the overall system design. At this stage you need to include drawings of a conceptual design for the IoT-based Smart utility monitoring system, not the implementation or prototype. This section

should be around 1200 words, excluding the figures/diagrams.

Task 2: Architecture Design Design a 3-layered architecture for your proposed model . Based on Task 1, apply a 3-layered architecture for your proposed IoT-based Smart Utility monitoring system using sensor technology and sensor networks to model a context aware and adaptive system. Make sure this 3-layered architecture includes the input, processing, and output layers. Include in your design the components (i.e., hardware or software units) necessary for each layer to achieve the desired functionality based on the specified requirements. In the explanation of your design, justify why you have included these components and explain how they perform the desired system functionalities. This section should be around 600 words, excluding the figures/diagrams.

Task 3: Data Acquisition Explain how you will get data from the physical sensors/devices deployed in the environment . Draw a diagram showing how to interface different I/O devices or components with Arduino microcontroller. Explain how to apply the sensors/devices used in the environment to supply the data to the system. This section should be around 400 words, excluding the figures/diagrams.

Task 4: System Implementation Describe and evaluate how a part of your system application can be implemented using a mobile device software API such as Android. Using a mobile device software API, such as Android, demonstrate how a part of your Smart Utility System application can be implemented and evaluate the design implications of using this device software and its overall efficacy. Please note , this does not mean implementing the application but relating your application design model to specific device APIs, to explain how the main part of your design could be implemented using the APIs / programming. This section should be around 500 words, excluding the figures/diagrams.

Task 5: Autonomous Learning-based System Convert your Smart Utility System into an autonomous learning-based system. The people living in Social Housing will have different lifestyles and health requirements. Your Smart Utility Design needs to explore how the system can learn and respond accordingly to the needs of the people living in the home. Think about how you can convert your Smart Utility System design into an autonomous system that can learn, a learning-based system. Provide a block diagram to show the autonomous learning-based system, then discuss and justify the changes to your proposed Smart Utility System design that will support autonomous operation in the real-life dynamic and diverse social housing environment. This section should be around 500 words, excluding the necessary figures/diagrams.

Submission Instructions Work through each of the tasks below. When you are done, upload all of the assessment documents to Canvas.

Policies

Success Criteria To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed. You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be awarded 50%.

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Course File Index

All files attached in this course. Open the link while logged in to download.

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